

Sets

Unit 1-2 Lecture Notes

MSDS 520: Math Fundamentals for Data Science
Meharry Medical College

These notes are the written companion to the Unit 1-2 slides and follow them section by section. They go into more detail than the slides and add many worked examples, most of them built on one small table of twelve patients (Section 1.3) so that every new idea is tried out on the same familiar data. Every exercise is followed by a worked solution; try each one before reading on.

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1 Why Sets?

1.1 A Problem to Motivate Us

Suppose someone asks you a simple-sounding question about a patient population:

*“How many patients have diabetes **or** hypertension, but **not both**?”*

Answering it carefully takes more vocabulary than it first appears. We need language for four things:

- a **collection** of patients (“the diabetics”),
- **combining** collections (“diabetic or hypertensive”),
- **removing** one collection from another (“but not both”),
- and **counting** what is left.

That language is **set theory**. It is the same logic as Unit 1-1 (AND, OR, NOT), but now applied to collections of things rather than to statements. In Unit 1-1 we asked whether a sentence is true; here we gather up all the things that make it true and work with that whole pile as a single object. We will answer the question above in Section 5.

Why is the question harder than it looks? Suppose a clinic tells you it has 40 diabetic patients and 55 hypertensive patients. It is tempting to add and say 95 patients have “diabetes or hypertension,” but that is wrong whenever some patients have both conditions: those patients were counted once among the diabetics and again among the hypertensives. And “but not both” asks us to throw exactly those double-counted patients out. Keeping track of who has been counted, and how many times, is precisely what sets and Venn diagrams are for.

1.2 Where This Shows Up in Data Science

Sets are everywhere once you look for them:

- **Cohorts and subgroups:** every “group of patients” is a set.
- **Joining data:** merging two tables is an intersection of their keys. The key is the identifying column, such as a patient ID, and the merged table keeps the IDs that appear in *both* tables.
- **Domains of functions:** the set of inputs a function accepts.
- **Reference ranges:** “normal” lab values are an **interval**, a set of numbers.
- **Probability:** an event is a set of outcomes (Unit 4).

To make these concrete, here is what each looks like in a typical health-data task.

- A **cohort** for a study of diabetic patients over 65 is the set of patients passing both tests; everyone else in the database is outside it.
- Merging a **lab-results table** with a **pharmacy table** on patient ID keeps the patients who appear in both: an intersection of two sets of IDs.
- The function “BMI = weight divided by height squared” only accepts a positive height; its **domain** is a set of allowed inputs.
- A fasting glucose “in the normal range” means the reading lies in a certain stretch of the number line, an **interval**; an *abnormal* reading lies outside it.
- “The patient tests positive” is, in Unit 4, an **event**: the set of all outcomes in which it happens.

Over the rest of these notes we build up the tools in this order:

1. **Sets:** membership, and **subsets**.

2. The **operations**: union, intersection, complement.
3. **Venn diagrams** and counting.
4. The standard **sets of numbers** and **intervals**.
5. **Ordered pairs** and tuples, where sets meet coordinates.

1.3 Our Running Example: A Small Study

Many of the worked examples in these notes use the same small, made-up study of twelve patients, labeled P1 to P12. For each patient we record an age, a sex, and four yes/no conditions.

Patient	Age	Sex	Diabetic	Hypertensive	Obese	Pregnant
P1	58	F	yes	yes	no	no
P2	71	M	yes	no	yes	no
P3	34	F	no	no	no	yes
P4	66	M	yes	yes	yes	no
P5	45	F	no	yes	no	no
P6	29	F	yes	no	no	yes
P7	80	M	no	yes	no	no
P8	52	M	yes	yes	no	no
P9	38	F	no	no	yes	no
P10	67	F	yes	no	yes	no
P11	41	M	no	no	no	no
P12	31	F	yes	yes	no	yes

This is a miniature version of what a real data table looks like: one **row** per patient, one **column** per recorded fact. Real studies have thousands of rows, which is exactly why we need notation that describes groups of patients by a *rule* rather than by pointing at rows one at a time. With twelve rows, though, we can check every answer by eye, which makes this table a good practice ground. Keep it handy; we will come back to it in almost every section.

2 Sets and Membership

2.1 What Is a Set?

Definition: Set

A **set** is a collection of objects, called its **elements** (or **members**).

The objects can be anything at all: numbers, letters, patients, even other sets. A few conventions govern how we write sets down.

- Sets are usually named with capital letters: A, B, C .

- Elements are listed inside curly braces, separated by commas:

$$A = \{1, 2, 3\}$$

- **Order does not matter:** $\{1, 2, 3\}$ and $\{3, 1, 2\}$ are the same set.
- **Repeats do not matter:** $\{1, 1, 2\}$ is just $\{1, 2\}$.

The last two rules come from a single idea: a set is defined entirely by *what is in it*, and nothing else. The only question a set answers is “is this object in me, or not?” Listing an element twice does not make it any more “in,” and listing the elements in a different order does not change which objects are in.

A good mental picture is a **guest list** at the door of an event. The list does not care in what order the names were written, and writing a name twice does not let that person in twice. All the list does is settle, for each person who walks up, “in” or “not in.”

Examples.

- The ABO blood types: $\{A, B, AB, O\}$, a set with four elements.
- The patients in our study (Section 1.3): $U = \{P1, P2, \dots, P12\}$. The “...” means “and so on, following the obvious pattern,” here P3 through P11.
- Reading down the “Diabetic” column and keeping the rows marked “yes” gives the set of diabetic patients,

$$D = \{P1, P2, P4, P6, P8, P10, P12\}.$$

In the same way, the hypertensive, obese, and pregnant patients are

$$H = \{P1, P4, P5, P7, P8, P12\},$$

$$O = \{P2, P4, P9, P10\},$$

$$G = \{P3, P6, P12\}.$$

(We use G , for “gravid,” the clinical word for pregnant, because P is already busy naming patients.) These four sets will appear again and again.

- A set can hold objects of different kinds, such as $\{7, \text{Tuesday}, P3\}$. That is legal, just not very useful.

Non-examples. Two things that look like sets but are not quite:

- **A vague description.** “The very sick patients” is not yet a set, because nobody can say for sure whether a particular patient is in or out. It becomes a set once a definite rule is agreed on, such as “patients admitted to the ICU this month.” A set must give a clear yes or no for every object.
- **A sequence of readings.** A patient’s fasting glucose on five mornings, 110, 95, 110, 102, 95, is not well described by a set. Order matters (did the readings go up or down?) and repeats matter (the value 110 happened twice). As a set it would collapse to $\{95, 102, 110\}$, losing both pieces of information. Section 8 introduces the right tool for ordered data, the *tuple*.

Worked example 2.1: Same set or not? Decide whether the two sets in each pair are equal. Remember that only membership counts.

1. $\{1, 2, 3\}$ and $\{3, 2, 1, 1\}$. The second list has a repeat and a different order, but it contains exactly the objects 1, 2, 3 and nothing else. **Equal.**
2. $\{a, b\}$ and $\{a, b, c\}$. The object c is in the second set but not the first. **Not equal.**
3. $\{O, A, O, B\}$ and $\{A, B, O\}$. Both hold exactly the three blood types A, B, O. **Equal.**
4. $\{1, \{1\}\}$ and $\{1\}$. The first set has two elements: the number 1, and the set $\{1\}$ (a box holding a 1). The second has only the number 1. **Not equal**, since the box $\{1\}$ is a different object from the number 1 (more on this in the next two subsections).

In each case the method was the same: list the distinct objects in each set, ignoring order and repeats, and compare the two lists.

Watch out: a set forgets order and repeats

Ten donors with blood types O, A, O, B, A, O, AB, O, A, O give the *set* of blood types $\{A, B, AB, O\}$: four elements, not ten. That is exactly what you want when the question is “which types occur?” and exactly wrong when the question is “how many donors are type O?” Before turning data into a set, ask whether you can afford to lose the order and the repeats.

This forgetting is also a tool. In Python, `set(["O", "A", "O", "B"])` produces the three distinct values "A", "B", "O" (displayed in some order, since a set has none). “Remove the duplicates” is one of the most common steps in cleaning data, and it is nothing more than turning a list into a set.

2.2 Membership

To say that an object is in a set, we use the symbol $\in$ (think “e” for “element”):

$$1 \in A \quad \text{“1 is an element of } A\text{”}$$

To say that it is not, we strike the symbol through:

$$5 \notin A \quad \text{“5 is not an element of } A\text{”}$$

Both lines use the set $A = \{1, 2, 3\}$ from above.

This connects directly to Unit 1-1. “ $x \in A$ ” is a **statement**: it is true or false, and so it can be combined with $\neg$, $\wedge$, $\vee$ like any other statement. For instance, $5 \notin A$ is just a shorter way of writing $\neg(5 \in A)$.

Notice that the truth of “ $x \in A$ ” depends on which x you pick: it is true for $x = 1$ and false for $x = 5$. That dependence is exactly what set-builder notation, later in this section, will use. Everything we do with sets will mirror something we did with logic.

Examples from the study. Reading the table in Section 1.3:

- $P3 \in G$: patient 3 is pregnant.
- $P1 \notin G$: patient 1 is not.
- $P7 \in H$ but $P7 \notin D$: patient 7 is hypertensive and not diabetic. Written with the connectives of Unit 1-1, $P7 \in H \wedge \neg(P7 \in D)$ is a true statement.

- $P11 \notin D$, $P11 \notin H$, $P11 \notin O$, and $P11 \notin G$: patient 11 has none of the four conditions.

Worked example 2.2: Membership in a set described by a rule. Let A be the set of all positive numbers. Decide whether each number is in A .

- 7: it is greater than 0, so $7 \in A$.
- -2 : it is less than 0, so $-2 \notin A$.
- 0.001: tiny, but still greater than 0, so $0.001 \in A$.
- 0: the test is “greater than 0,” and 0 is not greater than itself, so $0 \notin A$.

The last one is the kind of boundary case that trips people up. To decide membership, apply the rule *exactly as written*; do not go by a general feeling of what “positive” ought to include.

Watch out: elements can themselves be sets

Let $B = \{1, \{2, 3\}\}$. This set has exactly **two** elements: the number 1, and the set $\{2, 3\}$. So

$$1 \in B, \quad \{2, 3\} \in B, \quad \text{but} \quad 2 \notin B \quad \text{and} \quad 3 \notin B.$$

The 2 and the 3 are inside a box that is inside B ; they are not themselves elements of B . Membership only looks one level deep. (A real-life version: a hospital is a set of wards, and each ward is a set of patients. A patient is in a ward, and the ward is in the hospital, but the patient is not one of the hospital’s wards.)

2.3 Some Special Sets

Three kinds of sets come up often enough to get their own names.

Definition: Singleton

A set with exactly one element, such as $\{1\}$, is called a **singleton**.

Note that 1 and $\{1\}$ are different things: a number, and a box containing that number. It helps to picture the curly braces as the walls of a box. A box with one pen in it is still a box, not a pen, and in the same way $\{1\}$ is a set, not the number 1.

Definition: Empty set

The set with **no** elements is called the **empty set**:

$$\emptyset = \{ \}$$

The empty set is a box with nothing in it. It is what you get when a filter matches nobody, and even though it holds nothing, it is a perfectly good set.

Definition: Universal set

Often we fix a **universal set** U containing everything under discussion, e.g. all patients in the study.

Once U is fixed, “everything else” always means “everything else in U .” The universe draws the boundary of the conversation: every set we then talk about is made of things from U .

Examples of each.

- **Singletons.** In our study, the set of patients over 75 is $\{P7\}$: patient 7, aged 80, is the only one. The set of even prime numbers is $\{2\}$. A query that “finds exactly one match” returns a singleton.
- **The empty set.** The set of patients in our study who are both male and pregnant is $\emptyset$, and so is the set of months with 32 days. In a database, a query that returns zero rows has returned the empty set. That is a real, useful answer (“nobody matches”), not an error.
- **Universes.** For our study, $U = \{P1, \dots, P12\}$. For a hospital-wide question, U might be every patient in the electronic health record (EHR). For a question about numbers, U is often all real numbers (Section 6).

Worked example 2.3: The same description in two universes. “The patients who are not diabetic” depends on who “the patients” are. In our twelve-patient study it is the set $\{P3, P5, P7, P9, P11\}$, five patients: everyone in U except the seven in D . If U were instead the whole hospital EHR, the same words would describe thousands of people. Nothing about the phrase changed; only the universe did. This is why a careful analysis always states its universe first (“among adults admitted in 2025, ...”).

Watch out: $\emptyset$, $\{\emptyset\}$, 0, and $\{0\}$ are four different things

Object	What it is	Number of elements
0	the number zero (not a set at all)	not a set
$\emptyset$	an empty box	0
$\{0\}$	a box holding the number zero	1
$\{\emptyset\}$	a box holding an empty box	1

A box that contains an empty box is not empty: it has one thing in it. Databases show the same distinction. A counting query that returns one row holding the number 0 (“zero patients matched”) has returned something like $\{0\}$; a search that returns no rows at all has returned $\emptyset$.

2.4 Set-Builder Notation

Listing elements only works for small sets. For large or infinite ones, we describe the **rule** that members satisfy instead:

$$A = \{x \mid x > 0\} \quad \text{“the set of all positive numbers”}$$

Read the vertical bar (the “pipe”) as “**such that**”:

“the set of all x *such that* x is greater than 0.”

The part before the bar names a placeholder, x , which stands for a candidate member. The part after the bar is the test that a candidate must pass to get in.

Here are more examples.

- $\{x \mid x \text{ is a day of the week}\}$: “the set of all x such that x is a day of the week.”
- $\{x \mid x \text{ is a patient and } x \text{ is diabetic}\}$: the diabetic patients.
- $\{x \in U \mid x \text{ is diabetic}\}$: here membership is stated up front, before the bar. Read it as “the set of all x in U such that x is diabetic.” This is how the previous example is usually written once a universe of patients has been fixed: the “is a patient” part moves in front of the bar as $x \in U$.

The rule after the bar is a **statement about x** : logic again. For each candidate x the rule comes out true or false, and the set collects exactly those x that make it true.

Think of set-builder notation as a **filter**. The part before the bar says what kind of thing is being filtered (all numbers, all patients in U , all letters), and the part after the bar is the filter itself. Every candidate is run through the filter; the ones that come out “true” are the members. This is exactly what happens when you filter the rows of a data table: the table is the universe, the filter condition is the rule, and the rows that survive form the set.

There are two skills to practice: reading set-builder notation (turning a rule into a list, or into English), and writing it (turning a list or an English description into a rule).

Procedure: from set-builder notation to a list

1. Identify what x ranges over: the part before the bar, or the universe if one is fixed.
2. Run each candidate through the rule after the bar, one at a time.
3. Keep exactly the candidates that make the rule true.

Worked example 2.4: A small set of integers. The **integers** are the whole numbers together with their negatives: $\dots, -2, -1, 0, 1, 2, \dots$ (Section 6 gives them a symbol). List the set

$$\{x \mid x \text{ is an integer and } -2 \leq x < 3\}.$$

The candidates are the integers. The rule has two parts joined by AND: x must be at least -2 , and x must be less than 3. Going through the integers near that stretch: -3 fails the first part; -2 passes both (the $\leq$ allows equality); $-1, 0, 1, 2$ pass both; 3 fails the second part, since $3 < 3$ is false. So the set is

$$\{-2, -1, 0, 1, 2\}.$$

It has five elements. Notice how the two ends were treated differently: $\leq$ let -2 in, while $<$ kept 3 out.

Worked example 2.5: A rule involving squares. List the set

$$\{x \mid x \text{ is a positive integer and } x^2 < 20\}.$$

Test the positive integers in order: $1^2 = 1$, $2^2 = 4$, $3^2 = 9$, and $4^2 = 16$ are all less than 20, so $1, 2, 3, 4$ pass. Then $5^2 = 25$ is not less than 20, so 5 fails. We can stop here, because every larger integer has an even larger square and must fail too. The set is $\{1, 2, 3, 4\}$. This “stopping” step is typical: you cannot test infinitely many candidates one by one, so you look for a reason why all the remaining ones fail.

Worked example 2.6: A rule applied to the study table. With U the twelve patients of Section 1.3, list

$$\{x \in U \mid x \text{ is diabetic} \wedge x \text{ is over } 65\}.$$

A patient must pass both tests. It is efficient to take the first test first: the diabetics are P1, P2, P4, P6, P8, P10, P12. Now check each one's age: 58 (no), 71 (yes), 66 (yes), 29 (no), 52 (no), 67 (yes), 31 (no). The set is

$$\{P2, P4, P10\}.$$

Patient 7 is over 65 too (aged 80), but is not diabetic, so fails the first test and is left out. With AND, failing either test is enough to be excluded.

Worked example 2.7: The same rule with OR. Now list

$$\{x \in U \mid x \text{ is pregnant} \vee x \text{ is over } 65\}.$$

With OR, a patient gets in by passing *at least one* test. The pregnant patients are P3, P6, P12; the patients over 65 are P2, P4, P7, P10. Anyone on either list is in:

$$\{P2, P3, P4, P6, P7, P10, P12\}.$$

Seven patients. Changing AND to OR made the set bigger, and that is typical: AND is a stricter filter than OR.

Going the other way, from a list or a description to a rule, takes a little more thought, because you have to invent the rule.

Procedure: from English (or a list) to set-builder notation

1. Decide what x ranges over, and put it before the bar ($x \in U$, or “ x is an integer”).
2. Turn each condition in the description into a statement about x .
3. Join the statements with $\wedge$, $\vee$, $\neg$ exactly as the description demands.
4. Check: read your rule back in English, and test one element that should be in and one that should not.

Worked example 2.8: From a list to a rule. Write $\{1, 3, 5, 7, 9\}$ in set-builder notation. These are the odd numbers from 1 to 9, so

$$\{1, 3, 5, 7, 9\} = \{x \mid x \text{ is an odd integer and } 1 \leq x \leq 9\}.$$

Check: 5 is odd and between 1 and 9, so it is in; 11 is odd but too big, so it is out. There is more than one correct answer. For instance, $\{x \mid x \text{ is an odd integer and } 0 < x < 10\}$ describes exactly the same set, and so is equally right. Two rules are equivalent when they let in exactly the same objects.

Worked example 2.9: From an English description of patients. Write “the patients in the study younger than 40” in set-builder notation, and list it. The universe is the study, so $x \in U$ goes before the bar:

$$\{x \in U \mid x \text{ is younger than } 40\} = \{P3, P6, P9, P12\},$$

reading the ages 34, 29, 38, 31 from the table. Every other patient is 41 or older.

Worked example 2.10: A description with AND and NOT. Write “the women in the study who are hypertensive but not diabetic” in set-builder notation, and list it. There are three conditions: female, hypertensive, and *not* diabetic. The word “but” is logically just AND (it only adds a note of contrast), so

$$\{x \in U \mid x \text{ is female} \wedge x \text{ is hypertensive} \wedge \neg(x \text{ is diabetic})\}.$$

To list it, filter step by step. The hypertensive women are P1, P5, P12. Of these, P1 and P12 are diabetic, so they fail the last test. The set is the singleton {P5}.

Watch out: in everyday English, “and” sometimes means OR

“Send me the list of patients with diabetes and patients with hypertension” asks for *everyone in either group*: that is an OR, and the answer is the bigger set. “Send me the patients with diabetes and hypertension” asks for the patients who have *both*: an AND, and a smaller set. When you translate a request into a rule, ask of each “and”: must one patient satisfy both conditions, or is the speaker joining two lists? Getting this wrong is one of the most common ways a data pull returns the wrong cohort.

Watch out: the letter is only a placeholder

$\{x \mid x > 0\}$ and $\{t \mid t > 0\}$ are the same set: the letter just stands for “a candidate” and could be any symbol. What does matter is what the candidates range over. $\{x \mid x \text{ is an integer and } x > 0\} = \{1, 2, 3, \dots\}$ contains no fractions, while $\{x \mid x > 0\}$, taken over all numbers, contains 0.5 and π as well.

Exercise 2.1

Part 1. Translate each set from set-builder notation into plain English.

1. $\{x \mid x < 0\}$
2. $\{x \in U \mid x \text{ is pregnant}\}$, where U is all patients in the study.
3. $\{x \mid x \text{ is a month with exactly 30 days}\}$; then list its elements.

Part 2. Now go the other way: write each set in set-builder notation.

1. The set of all patients who are diabetic **and** over 65.
2. The set of all letters that are vowels.
3. The set of all numbers between 10 and 20.

Solution to Exercise 2.1

Part 1, into English. Read the bar as “such that” and the rule after it as an ordinary sentence.

1. “The set of all x *such that* x is less than 0”: the negative numbers.
2. “The set of all patients in the study who are pregnant.” The $x \in U$ in front of the bar is what says “patients in the study.”

3. “The set of all months with exactly 30 days.” Listing it:

$\{\text{April, June, September, November}\}.$

Part 2, into set-builder. Taking U to be the patients:

1. $\{x \in U \mid x \text{ is diabetic} \wedge x \text{ is over } 65\}$. The “and” is the AND ($\wedge$) from Unit 1-1: a patient must pass **both** tests to get in.
2. $\{x \mid x \text{ is a vowel}\} = \{a, e, i, o, u\}$. This set is small enough to list as well.
3. $\{x \mid 10 < x < 20\}$.

That last one is slippery: “between” never says whether 10 and 20 themselves count. We left them out here, but $\{x \mid 10 \leq x \leq 20\}$ is an equally fair reading of the English. **Intervals** (Section 7) will give us notation that pins this down.

2.5 Cardinality

Definition: Cardinality

The **cardinality** of a set A , written $|A|$, is the number of elements it contains.

The vertical bars here are not absolute value: wrapped around a set, they mean “how many elements.” Some examples:

- $A = \{1, 2, 3\} \Rightarrow |A| = 3$.
- $|\{7\}| = 1$: every singleton has cardinality 1.
- $|\emptyset| = 0$: the empty set has nothing in it to count.

Because repeats do not matter in a set, cardinality counts *distinct* elements. Every “how many patients ...” question is a cardinality question, and we will come back to this when we count with Venn diagrams in Section 5.

In our study, $|U| = 12$, $|D| = 7$, $|H| = 6$, $|O| = 4$, and $|G| = 3$. The set of months with exactly 31 days (January, March, May, July, August, October, December) has cardinality 7.

Worked example 2.11: Rows versus patients. A clinic’s visit log for one week lists the patient seen at each of eight visits:

P3, P7, P3, P1, P9, P7, P3, P12.

How many different patients were seen? Collect the visits into a set, and the repeats disappear:

$\{P1, P3, P7, P9, P12\}$, with cardinality 5.

So the log has 8 rows but only 5 patients: patient 3 came three times and patient 7 twice. In data work this is the difference between counting rows and counting *distinct* values, and confusing the two is a classic reporting error. (In Python, `len(set(visits))`; in a database query, `COUNT(DISTINCT patient_id)`.)

Worked example 2.12: Letters of a word. Find the cardinality

$$|\{x \mid x \text{ is a letter in the word "mississippi"}\}|.$$

The word has eleven letters, but only four *different* ones appear: m, i, s, p. So the set is $\{m, i, s, p\}$ and its cardinality is 4. A quick way to do this by hand is to read the word left to right and write down each letter only the first time you meet it.

Worked example 2.13: Sets inside sets. Find $|\{1, \{2, 3\}\}|$ and $|\{\emptyset\}|$. The first set has two elements, the number 1 and the box $\{2, 3\}$, so its cardinality is 2, not 3: a box counts as one element no matter what is inside it. The second set has one element, the empty box, so $|\{\emptyset\}| = 1$, even though $|\emptyset| = 0$.

Watch out: count distinct elements, one level deep

Two errors cover most wrong cardinalities. Counting a repeated element more than once ($|\{1, 1, 2\}|$ is 2, not 3), and counting the contents of an inner set instead of the set itself ($|\{1, \{2, 3\}\}|$ is 2, not 3).

Exercise 2.2

Let

$$A = \{2, 4, 6, 8\}$$

$$B = \{x \mid x \text{ is a letter in the word "level"}\}$$

1. Write B by listing its elements.
2. Find $|A|$ and $|B|$.
3. True or false: $4 \in A$, $\{4\} \in A$, $5 \notin A$.

Solution to Exercise 2.2

1. Listing B . The letters of "level," in order, are l, e, v, e, l. But **repeats do not matter**, so each distinct letter appears once:

$$B = \{l, e, v\}.$$

2. Cardinality. $|A| = 4$ and $|B| = 3$: three *distinct* letters, not five. Counting all five letters of the word would count the repeats, and a set has none.

3. True or false.

- $4 \in A$ is **true**: 4 is one of the listed elements.
- $\{4\} \in A$ is **false**. The elements of A are numbers; $\{4\}$ is a *set* (a box holding 4), and no set appears among the elements of A .
- $5 \notin A$ is **true**: 5 is not listed.

The second one is the trap. Membership asks about the exact object, and the box $\{4\}$ is a different object from the number 4.

2.6 Extra Practice

Exercise 2.3

List each set, then give its cardinality.

1. $\{x \mid x \text{ is an integer and } -3 < x \leq 2\}$
2. $\{x \mid x \text{ is a letter in the word "banana"}\}$
3. $\{1, 2, 2, 3, 3, 3\}$
4. $\{\emptyset, \{\emptyset\}\}$

Solution to Exercise 2.3

1. The rule lets in integers strictly greater than -3 and at most 2 . So -3 is out (the $<$ excludes it), 2 is in (the $\leq$ includes it), and the set is $\{-2, -1, 0, 1, 2\}$, with cardinality 5 .
2. The letters of "banana" are b, a, n, a, n, a, but only three are distinct: $\{b, a, n\}$, cardinality 3 .
3. Repeats do not count: $\{1, 2, 2, 3, 3, 3\} = \{1, 2, 3\}$, cardinality 3 .
4. Two different objects are listed: the empty box $\emptyset$, and a box $\{\emptyset\}$ holding an empty box. So the cardinality is 2 . It does not matter that neither element has anything "real" inside it; each is one object.

Exercise 2.4

Using the study table in Section 1.3, write each set in set-builder notation and then list it.

1. The patients aged 40 to 60, inclusive.
2. The male patients who are obese.
3. The patients who are not hypertensive.

Solution to Exercise 2.4

In each case the universe is the study, so the rule starts with $x \in U$.

1. $\{x \in U \mid 40 \leq \text{age of } x \leq 60\}$. "Inclusive" tells us to use $\leq$ at both ends, so a patient aged exactly 40 or exactly 60 would count. Scanning the ages: 58, 45, 52, 41 qualify, giving $\{P1, P5, P8, P11\}$.
2. $\{x \in U \mid x \text{ is male} \wedge x \text{ is obese}\}$. The men are P2, P4, P7, P8, P11; of these only P2 and P4 are obese. The set is $\{P2, P4\}$.
3. $\{x \in U \mid \neg(x \text{ is hypertensive})\}$. Everyone in U except the six members of H : $\{P2, P3, P6, P9, P10, P11\}$. As a check, $6 + 6 = 12$: every patient is either hypertensive or not, never both.

Exercise 2.5

Let $C = \{a, \{b\}, \emptyset\}$. True or false?

1. $a \in C$
2. $b \in C$
3. $\{b\} \in C$
4. $\emptyset \in C$
5. $|C| = 3$

Solution to Exercise 2.5

List the elements of C first: they are a , the box $\{b\}$, and the empty box $\emptyset$.

1. $a \in C$ is **true**: a is listed.
2. $b \in C$ is **false**: b sits inside the box $\{b\}$, one level down, and is not itself an element of C .
3. $\{b\} \in C$ is **true**: the box is listed.
4. $\emptyset \in C$ is **true**, and this is the surprise. The empty set is *not* an element of every set (Exercise 3.1 below makes that point), but here it is written in the list, so it is an element of this particular set.
5. $|C| = 3$ is **true**: three distinct objects are listed.

3 Subsets

3.1 What Is a Subset?

Definition: Subset

A is a **subset** of B , written $A \subseteq B$, if **every** element of A is also an element of B .

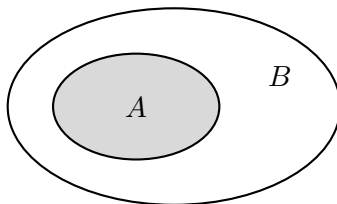
Written with the implication from Unit 1-1, the condition is that for every x ,

$$x \in A \rightarrow x \in B.$$

In words: if x is in A , then x is in B . The only way to break this promise is to find an x that is in A but not in B , which is exactly the one false row of the implication truth table.

For example, $\{1, 2\} \subseteq \{1, 2, 3\}$, and also $\{1, 2, 3\} \subseteq \{1, 2, 3\}$: a set is a subset of itself. The second one surprises people, but check it against the definition. Every element of $\{1, 2, 3\}$ is certainly an element of $\{1, 2, 3\}$, and nothing in the definition requires B to have anything extra.

In a picture, $A \subseteq B$ means the region for A sits entirely inside the region for B :



Keep the two symbols apart: $\in$ relates an *element* to a set, while $\subseteq$ relates a *set* to a set. So $1 \in \{1, 2\}$, but $\{1\} \subseteq \{1, 2\}$.

In plain words, $A \subseteq B$ says “ A is contained in B ”: there is nothing in A that B lacks. Subsets arise every time you **filter**. Keeping only some rows of a table can never produce a row that was not there to begin with, so the rows you keep always form a subset of the rows you started with.

When we write “ $A \not\subseteq B$ ” (with a stroke through the symbol, just as for $\notin$), we mean A is *not* a subset of B : at least one element of A is missing from B .

Procedure: checking whether $A \subseteq B$

1. Go through the elements of A one at a time, and ask of each: is it in B ?
2. The moment one answer is “no,” stop: $A \not\subseteq B$, and that element is your **counterexample**.
3. If every element of A passes, then $A \subseteq B$.
4. If A is infinite, you cannot check one at a time. Instead take an arbitrary element of A , knowing only the rule it satisfies, and show it must satisfy B ’s rule too.

Step 2 is the implication from Unit 1-1 at work: “ $x \in A \rightarrow x \in B$ ” fails only for an x with $x \in A$ true and $x \in B$ false, and a single such x is enough to break the promise.

Worked example 3.1: A subset of a small set. Is $\{2, 4\} \subseteq \{1, 2, 3, 4, 5\}$? Check each element of the left-hand set: 2 is in $\{1, 2, 3, 4, 5\}$, and so is 4. Every element passed, so **yes**. Notice that we never looked at 1, 3, or 5. The extra elements of the bigger set are irrelevant; only the smaller set’s elements get tested.

Worked example 3.2: Not a subset. Is $\{1, 3, 6\} \subseteq \{1, 2, 3, 4, 5\}$? Check: 1 is in, 3 is in, 6 is *not*. So **no**: $\{1, 3, 6\} \not\subseteq \{1, 2, 3, 4, 5\}$, with counterexample 6. Two out of three passing does not help; “every” means every.

Worked example 3.3: Subsets in the study table. Using the sets from Section 1.3:

- **Is $G \subseteq F$,** where F is the set of female patients? The pregnant patients are P3, P6, P12, and the table lists all three as F. So **yes**, every pregnant patient is female, as it must be. (This is also a useful data-quality check: a record showing a pregnant male patient would signal an entry error.)
- **Is $D \subseteq H$,** that is, is every diabetic also hypertensive? Check the diabetics in order: P1 is hypertensive, but P2 is not. Stop: **no**, and P2 is a counterexample. (P6 and P10 would also serve, but one is enough.)
- **Is $\{P4, P8\} \subseteq H$?** Both patients are hypertensive, so **yes**.

Worked example 3.4: A subset of an infinite set. Let $A = \{x \mid x \text{ is a multiple of } 4\}$ and $B = \{x \mid x \text{ is an even integer}\}$, where the multiples of 4 are the integers $\dots, -8, -4, 0, 4, 8, 12, \dots$. Is $A \subseteq B$? There are infinitely many elements to check, so use step 4. Take an arbitrary $x \in A$. Being a multiple of 4 means $x = 4k$ for some integer k . Then

$$x = 4k = 2 \cdot (2k),$$

which is 2 times an integer, so x is even and $x \in B$. Since x was arbitrary, this works for every element of A : **yes**, $A \subseteq B$.

What about the other direction, $B \subseteq A$? Try an element: 2 is even, so $2 \in B$, but 2 is not a multiple of 4. So $B \not\subseteq A$, with counterexample 2. A single example can *disprove* a containment; proving one for an infinite set takes an argument about an arbitrary element.

Watch out: $\in$ versus $\subseteq$, all in one place

Let $A = \{1, 2, \{3\}\}$, a set with three elements: 1, 2, and the box $\{3\}$.

Statement	T/F	Why
$1 \in A$	T	1 is listed
$\{1\} \in A$	F	the box $\{1\}$ is not listed
$\{1\} \subseteq A$	T	its only element, 1, is in A
$3 \in A$	F	3 sits inside the box $\{3\}$, one level down
$\{3\} \in A$	T	the box $\{3\}$ is listed
$\{3\} \subseteq A$	F	its only element, 3, is not in A
$\{\{3\}\} \subseteq A$	T	its only element, the box $\{3\}$, is in A

The test for each symbol never changes. For “thing $\in A$,” look for the thing itself in A ’s list. For “ $S \subseteq A$,” take each element of S and look for *it* in A ’s list.

3.2 Proper Subsets, and the Empty Set**Definition: Proper subset**

A is a **proper** subset of B , written $A \subset B$, if $A \subseteq B$ **and** $A \neq B$. That is, B has at least one element that A does not.

$A \subseteq B$	contained in, possibly equal
$A \subset B$	contained in, definitely smaller

The two symbols work like $\leq$ and $<$ for numbers: the extra line under $\subseteq$ plays the same role as the line under $\leq$, allowing the two sides to be equal.

Fact: $\emptyset \subseteq B$ for every set B

The empty set is a subset of **everything**.

Why? To fail to be a subset of B , $\emptyset$ would need an element that is not in B , and it has no elements at all to offer. In the language of Unit 1-1, the condition “ $x \in \emptyset \rightarrow x \in B$ ” has an antecedent that is false for every x , so the implication is **vacuously true**. Nothing extra has to be agreed on here: the fact follows straight from the definition of a subset.

Examples.

- $\{1, 2\} \subset \{1, 2, 3\}$: contained, and 3 is extra. A proper subset.
- $\{1, 2, 3\} \subseteq \{3, 2, 1\}$, but *not* $\{1, 2, 3\} \subset \{3, 2, 1\}$: the two are the same set, so nothing is extra.
- In the study, $G \subset F$: every pregnant patient is female, and F also contains women who are not pregnant (patient 1, for one). Likewise $\{P2, P4, P10\} \subset D$: the diabetics over 65 are some, but not all, of the diabetics.
- $\emptyset \subset \{7\}$: the empty set is a proper subset of any set that has at least one element. The only set of which $\emptyset$ is *not* a proper subset is $\emptyset$ itself.

Worked example 3.5: A vacuous subset in the study. Let

$$V = \{x \in U \mid x \text{ is older than } 100\}.$$

Nobody in the study is over 100, so $V = \emptyset$. Is $V \subseteq D$? By the fact above, yes: V has no member who could fail to be diabetic. In words, “every patient in the study over 100 is diabetic” is true, and so is “every patient in the study over 100 is *not* diabetic.” Both are vacuously true, because there is no such patient to test. This can feel like a trick, but it is the only answer consistent with the definition, and it matters in practice: a filter that matches nobody still produces a perfectly good (empty) subset of every other set.

Watch out: not every book uses $\subset$ the same way

In these notes $A \subset B$ means a *proper* subset, and $A \subseteq B$ allows equality. Some textbooks and software documentation instead write $\subset$ for “subset, possibly equal,” and $\subsetneq$ when they mean proper. When you read another source, check which convention it uses before trusting a statement like “ $A \subset A$.”

3.3 Set Equality

Definition: Set equality

Two sets are equal when each contains the other:

$$A = B \quad \text{means} \quad A \subseteq B \quad \text{and} \quad B \subseteq A$$

Unpack the two halves. $A \subseteq B$ says everything in A is also in B ; $B \subseteq A$ says everything in B is also in A . Together they say the two sets have exactly the same elements, and since a set is defined entirely by what is in it, that makes them the same set.

This is the set-theory version of the bi-implication $\leftrightarrow$ from Unit 1-1: $x \in A$ if and only if $x \in B$. Just as $p \leftrightarrow q$ is $(p \rightarrow q) \wedge (q \rightarrow p)$, set equality is two subset statements joined by AND.

Proof strategy: showing two sets are equal

To prove two sets are the same, prove **both** containments:

1. Take an arbitrary $x \in A$ and show $x \in B$.
2. Take an arbitrary $x \in B$ and show $x \in A$.

“Arbitrary” means we assume nothing about x except that it belongs to the set, so whatever we show about it holds for every element at once.

Why bother with two directions? Because each direction catches a different kind of mismatch. Showing $A \subseteq B$ rules out elements of A that are missing from B ; showing $B \subseteq A$ rules out elements of B that are missing from A . Checking only one direction is like checking that every name on your list is on mine, but not the other way around: my list could still have extra names.

Worked example 3.6: Two words with the same letters. Let A be the set of letters in “listen” and B the set of letters in “silent.” Is $A = B$?

- $A \subseteq B$: the letters of “listen” are l, i, s, t, e, n, and each of them appears in “silent.”

- $B \subseteq A$: the letters of “silent” are s, i, l, e, n, t, and each appears in “listen.”

Both containments hold, so $A = B = \{e, i, l, n, s, t\}$. The two words are different (order matters for words), but their *sets* of letters are equal (order does not matter for sets).

Worked example 3.7: A set described by an equation. Show that

$$\{x \mid x \text{ is a number with } x^2 = 4\} = \{-2, 2\}.$$

Call the left-hand set A and the right-hand set B .

- $B \subseteq A$: check the two elements of B . $(-2)^2 = 4$ and $2^2 = 4$, so both satisfy A ’s rule.
- $A \subseteq B$: take an arbitrary x with $x^2 = 4$. The only numbers whose square is 4 are 2 and -2 (a number bigger than 2 in size has a square bigger than 4, and one smaller in size has a square smaller than 4). So x is 2 or -2 , that is, $x \in B$.

Both directions hold, so the sets are equal. The direction that people forget is $B \subseteq A$, and it is the one that catches a *missed* solution: if we had claimed the answer was $\{2\}$, the check “does -2 satisfy the rule?” would expose it.

Worked example 3.8: Two rules that describe the same infinite set. Working with integers, show that

$$\{x \mid x \text{ is even and a multiple of } 3\} = \{x \mid x \text{ is a multiple of } 6\}.$$

Call the left-hand set A and the right-hand set B . Both are infinite, so each direction needs an argument about an arbitrary element.

- $B \subseteq A$: take any $x \in B$, so $x = 6k$ for some integer k . Then $x = 2 \cdot (3k)$ is even, and $x = 3 \cdot (2k)$ is a multiple of 3. So $x \in A$.
- $A \subseteq B$: take any $x \in A$. Since x is a multiple of 3, $x = 3m$ for some integer m . Could m be odd? Then $3m$ would be odd times odd, which is odd; but x is even. So m must be even, say $m = 2k$, and then $x = 3 \cdot 2k = 6k$. So $x \in B$.

Both containments hold, so $A = B$. This is a small version of something data scientists do often: two different-looking filters that are secretly the same, where proving they agree lets you replace a complicated rule with a simpler one.

A non-example. The multiples of 4 and the even integers are *not* equal. The previous subsection showed that one containment holds (every multiple of 4 is even) but the other fails (the even number 2 is not a multiple of 4). Equality needs both.

Watch out: checking a few examples is not a proof

For finite sets you can check every element. For infinite sets, testing 6, 12, and 18 and seeing that each is even and a multiple of 3 shows that the rule *seems* right, but it cannot rule out a failure further along. Examples can disprove a containment (one counterexample is enough) but can never prove one about infinitely many elements; for that you need an argument about an arbitrary element.

Exercise 3.1

Let $A = \{1, 2, 3\}$.

1. List **all** subsets of A . How many are there?
2. Which of those are **proper** subsets?
3. True or false: $\emptyset \subseteq A$, $\emptyset \in A$, $A \subseteq A$, $A \subset A$.

Solution to Exercise 3.1

1. **All subsets.** Organize them by size, so none gets missed:

Size	Subsets	How many
0	$\emptyset$	1
1	$\{1\}, \{2\}, \{3\}$	3
2	$\{1, 2\}, \{1, 3\}, \{2, 3\}$	3
3	$\{1, 2, 3\}$	1

That is $1 + 3 + 3 + 1 = 8$ subsets. In general, a set with n elements has 2^n subsets, here $2^3 = 8$; the next subsection explains why.

2. **Proper subsets.** Every one of them except $\{1, 2, 3\}$ itself: **seven**. Note that $\emptyset$ counts, since $\emptyset \subseteq A$ and $\emptyset \neq A$.

3. **True or false.**

- $\emptyset \subseteq A$ is **true**. The empty set is a subset of everything.
- $\emptyset \in A$ is **false**. The elements of A are 1, 2, 3, and $\emptyset$ is not one of them. The empty set is a *subset* of every set, but an *element* only of a set that lists it.
- $A \subseteq A$ is **true**: every set is a subset of itself.
- $A \subset A$ is **false**: a proper subset demands $A \neq A$, which can never hold.

3.4 The Power Set

The eight subsets we just listed are objects in their own right, so we can collect them into a set.

Definition: Power set

The **power set** of A , written $\mathcal{P}(A)$, is the set of **all** subsets of A :

$$\mathcal{P}(A) = \{ S \mid S \subseteq A \}$$

Its elements are themselves **sets**, not the original elements.

For $A = \{1, 2, 3\}$,

$$\mathcal{P}(A) = \{ \emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\} \}.$$

So $|\mathcal{P}(A)| = 8$, and both $\emptyset$ and A itself are members.

Why 2^n ? To build a subset, go through the elements of A one at a time and make one yes/no choice for each: in, or out. With 3 elements there are $2 \times 2 \times 2 = 8$ ways to make the choices, and

each way gives a different subset. For example, in, out, in gives $\{1, 3\}$, and out, out, out gives $\emptyset$. This is the same count as the rows of an n -variable truth table (Unit 1-1), and for the same reason: each row there is one T/F choice per statement.

Watch the symbols: $\{1\} \subseteq A$, but $\{1\} \in \mathcal{P}(A)$. The set $\{1\}$ is a subset of A , because its one element is in A . That is exactly what makes it an *element* of the power set, whose elements are the subsets of A .

The in/out choices can be laid out exactly like a truth table, one row per subset. For $\{a, b, c\}$:

a	b	c	Subset
in	in	in	$\{a, b, c\}$
in	in	out	$\{a, b\}$
in	out	in	$\{a, c\}$
in	out	out	$\{a\}$
out	in	in	$\{b, c\}$
out	in	out	$\{b\}$
out	out	in	$\{c\}$
out	out	out	$\emptyset$

Replace “in” by T and “out” by F and this is the left half of a three-statement truth table from Unit 1-1, row for row. Every subset appears exactly once, because every pattern of choices appears exactly once.

Small power sets. Working up from nothing:

$$\begin{array}{ll}
 \mathcal{P}(\emptyset) = \{\emptyset\} & \text{one subset: } 2^0 = 1 \\
 \mathcal{P}(\{a\}) = \{\emptyset, \{a\}\} & \text{two subsets: } 2^1 = 2 \\
 \mathcal{P}(\{a, b\}) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\} & \text{four subsets: } 2^2 = 4
 \end{array}$$

Each new element doubles the count, because every old subset now comes in two versions: without the new element, and with it. That doubling is what makes the count 2^n : 1, 2, 4, 8, 16, 32, ... for $n = 0, 1, 2, 3, 4, 5, \dots$

Worked example 3.9: Ordering a lab panel. A clinician can order any combination of four tests: glucose, cholesterol, creatinine, and hemoglobin. How many different orders are possible, counting “order nothing” as one option? Each possible order is a subset of the set of four tests, so the answer is the number of subsets:

$$|\mathcal{P}(\{\text{glucose, cholesterol, creatinine, hemoglobin}\})| = 2^4 = 16.$$

Organized by size, as in Exercise 3.1: 1 way to order no tests, 4 ways to order exactly one, 6 ways to order exactly two, 4 ways to order exactly three, and 1 way to order all four. Indeed $1 + 4 + 6 + 4 + 1 = 16$. If “order nothing” does not count, there are $16 - 1 = 15$ possible orders.

Worked example 3.10: Why trying every subset of features does not scale. In machine learning, a model is built from some chosen set of input variables, called *features*. Suppose a study has 10 candidate features and we want the best model among all possible feature choices. Every choice is a subset, so there are $2^{10} = 1024$ of them, counting the empty choice. With 30 candidate

features there would be 2^{30} , more than a billion. The number of subsets grows so fast that “try them all” stops being possible very quickly, which is why data scientists use cleverer selection methods. The power set is the reason.

Watch out: 2^n , not $2n$, and $\mathcal{P}(\emptyset)$ is not empty

A set with 10 elements has $2^{10} = 1024$ subsets, not $2 \times 10 = 20$. The count doubles with each element; it does not go up by two. And the power set of the empty set is $\{\emptyset\}$, a set with *one* element (the empty set is a subset of itself), so $|\mathcal{P}(\emptyset)| = 2^0 = 1$, not 0.

3.5 Extra Practice

Exercise 3.2

Let $B = \{x, y\}$.

1. List $\mathcal{P}(B)$ and find $|\mathcal{P}(B)|$.
2. Which elements of $\mathcal{P}(B)$ are proper subsets of B ?
3. True or false: $\{x\} \in \mathcal{P}(B)$, $x \in \mathcal{P}(B)$, $\emptyset \in \mathcal{P}(B)$, $\emptyset \subseteq \mathcal{P}(B)$.

Solution to Exercise 3.2

1. One in/out choice for each of the two elements gives four subsets:

$$\mathcal{P}(B) = \{\emptyset, \{x\}, \{y\}, \{x, y\}\}, \quad |\mathcal{P}(B)| = 2^2 = 4.$$

2. All of them except $B = \{x, y\}$ itself: $\emptyset$, $\{x\}$, and $\{y\}$.
3. $\{x\} \in \mathcal{P}(B)$ is **true**: $\{x\}$ is a subset of B , so it is one of the elements of the power set. $x \in \mathcal{P}(B)$ is **false**: the elements of $\mathcal{P}(B)$ are *sets*, and the letter x on its own is not one of them. $\emptyset \in \mathcal{P}(B)$ is **true**: the empty set is a subset of B , so it is listed in the power set. And $\emptyset \subseteq \mathcal{P}(B)$ is **true** as well, since the empty set is a subset of every set, including this one. The empty set is both an element and a subset here, for two different reasons.

Exercise 3.3

A set S has 5 elements. How many subsets does it have? How many proper subsets? How many subsets are neither empty nor all of S ?

Solution to Exercise 3.3

The number of subsets is $2^5 = 32$. A proper subset is any subset except S itself, so there are $32 - 1 = 31$. Removing both S and $\emptyset$ leaves $32 - 2 = 30$. In a study with five candidate risk factors, 30 is the number of ways to choose “some, but not none and not all” of them.

Exercise 3.4

Using the study table in Section 1.3, decide whether each statement is true or false; give a counterexample for each false one.

1. $H \subseteq D$
2. $\{P2, P5\} \subseteq D$
3. $\{P3, P12\} \subseteq G$
4. $G \subset G$
5. $\emptyset \subseteq O$

Solution to Exercise 3.4

1. **False.** Patient 5 is hypertensive but not diabetic (P7 also works).
2. **False.** P2 is diabetic, but P5 is not, so the counterexample is P5.
3. **True.** Both are pregnant, so both are in G .
4. **False.** A proper subset must differ from the set, and $G = G$. (Here the counterexample is not an element: the claim fails because the “not equal” half of the definition fails.)
5. **True,** vacuously: $\emptyset$ has no elements that could be missing from O .

Exercise 3.5

Working with integers, show that $\{x \mid x \text{ is a multiple of } 10\} \subseteq \{x \mid x \text{ is a multiple of } 5\}$, and show that the reverse containment fails.

Solution to Exercise 3.5

The containment. Take an arbitrary multiple of 10, say $x = 10k$ for some integer k . Then $x = 5 \cdot (2k)$, which is 5 times an integer, so x is a multiple of 5. Since x was arbitrary, every multiple of 10 is a multiple of 5.

The reverse fails. One counterexample settles it: 5 is a multiple of 5 but not of 10. So the multiples of 5 are not a subset of the multiples of 10, and the two sets are not equal. In the language of this section, the multiples of 10 form a *proper* subset of the multiples of 5.

4 Set Operations

Now we combine sets to make new ones. Each operation in this section is one of the logical connectives from Unit 1-1, applied to the statement “ x is in the set.”

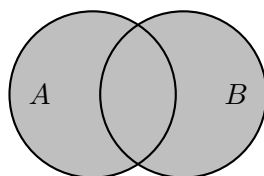
4.1 Union

Definition: Union

The **union** of A and B , written $A \cup B$, is the set of everything in A , in B , or in both:

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

In a picture, the union is everything inside either circle:



Union is the set version of the logical **OR**, and it is the *inclusive* OR: elements in both sets are included, listed once. For example,

$$\{1, 2, 3\} \cup \{2, 3, 4\} = \{1, 2, 3, 4\}.$$

The 2 and the 3 are in both sets, but a set has no repeats, so each appears only once in the union.

How to compute a union by hand. Write down every element of the first set, then go through the second set and add each element that is not already written. For $\{1, 2, 3\} \cup \{2, 3, 4\}$: write 1, 2, 3; then 2 is already there, 3 is already there, and 4 is new, so add it.

More examples.

- $\{a, b\} \cup \{b, c\} = \{a, b, c\}$.
- $\{1, 2\} \cup \emptyset = \{1, 2\}$: adding nothing changes nothing.
- $\{1, 2\} \cup \{1, 2\} = \{1, 2\}$: the union of a set with itself is itself.
- In the study, the patients who are diabetic *or* hypertensive:

$$D \cup H = \{P1, P2, P4, P5, P6, P7, P8, P10, P12\}.$$

Start from the seven diabetics and add the hypertensives not already listed, P5 and P7. The result has 9 patients.

In data work, a union is what you get when you **stack two lists and remove duplicates**. If one registry lists the patients with a diabetes diagnosis and another lists the patients on insulin, the union is everyone who appears on either list, each patient once.

Watch out: a union is not a sum

In the study, $|D| = 7$ and $|H| = 6$, but $|D \cup H| = 9$, not 13. The four patients who are in both sets (P1, P4, P8, P12) appear only once in the union. Adding the sizes of two sets overcounts whatever they share; Section 5 turns this observation into a formula.

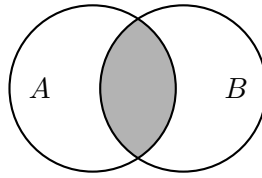
4.2 Intersection

Definition: Intersection

The **intersection** of A and B , written $A \cap B$, is the set of everything in **both**:

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

In a picture, the intersection is only the overlap of the two circles:



Intersection is the set version of the logical **AND**. For example,

$$\{1, 2, 3\} \cap \{2, 3, 4\} = \{2, 3\}.$$

If $A \cap B = \emptyset$, the sets share nothing and are called **disjoint**. In a picture, disjoint sets are two circles that do not overlap at all.

How to compute an intersection by hand. Go through the elements of one set, and keep each one that also appears in the other. For $\{1, 2, 3\} \cap \{2, 3, 4\}$: 1 is not in the second set (drop it), 2 is (keep), 3 is (keep).

More examples.

- In the study, the patients who are diabetic *and* hypertensive: go through D and keep the patients also in H ,

$$D \cap H = \{P1, P4, P8, P12\}.$$

- The pregnant patients and the male patients are disjoint: $G \cap M = \emptyset$, where M is the set of men.
- The even integers and the odd integers are disjoint: no integer is both.
- $\{1, 2\} \cap \emptyset = \emptyset$: nothing can be in both a set and the empty set.

Worked example 4.1: An intersection is an inner join. A lab-results table contains patients $\{P1, P2, P4, P7, P9\}$, and a pharmacy table contains patients $\{P2, P3, P4, P9, P11\}$. An analyst wants to study lab values against prescriptions, so needs patients present in *both* tables. Keep each lab-table patient who also appears in the pharmacy table: P1 no, P2 yes, P4 yes, P7 no, P9 yes. The merged data covers

$$\{P1, P2, P4, P7, P9\} \cap \{P2, P3, P4, P9, P11\} = \{P2, P4, P9\}.$$

In database language this is an *inner join* on patient ID. It explains a common surprise: merging two tables of five patients each can leave only three. The patients who fell out were never in both.

Watch out: intersection makes things smaller, union makes things bigger

$A \cap B$ is always contained in both A and B , while $A \cup B$ always contains both. If you intersect two sets and get something *bigger* than either one, or take a union and get something smaller, you have swapped the two operations. The logic is the same as in Unit 1-1: AND is harder to satisfy than OR.

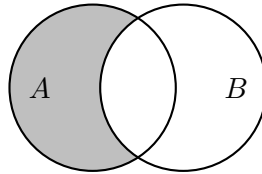
4.3 Difference (Relative Complement)

Definition: Set difference

The **difference** $A \setminus B$ is everything in A that is **not** in B :

$$A \setminus B = \{x \in A \mid x \notin B\}$$

In a picture, start with A and cut away the part that B covers:



Difference is the set version of **AND NOT**: $x \in A \setminus B$ means $x \in A \wedge \neg(x \in B)$. The name “relative complement” says the same thing: it is everything outside B , but only counting what lies inside A .

Unlike union and intersection, **order matters**:

$$\{1, 2, 3\} \setminus \{2, 3, 4\} = \{1\}$$

$$\{2, 3, 4\} \setminus \{1, 2, 3\} = \{4\}$$

Swapping the two sets changes which one you are removing from, and so changes the answer.

How to compute a difference by hand. Go through the elements of the *first* set, and cross out each one that appears in the second. Elements of the second set that are not in the first play no part at all: there is nothing to cross out.

More examples.

- In the study, the diabetics who are not hypertensive, and the other way around:

$$D \setminus H = \{P2, P6, P10\}, \quad H \setminus D = \{P5, P7\}.$$

For $D \setminus H$, start from the seven diabetics and cross out the four who are also hypertensive.

- When $A \subseteq B$, everything in A gets crossed out, so $A \setminus B = \emptyset$. For example, $G \setminus F = \emptyset$: no pregnant patient is outside the set of women.
- When A and B are disjoint, nothing gets crossed out, so $A \setminus B = A$. For example, $G \setminus M = G$.

In data work, a difference answers “who is on this list but missing from that one?” For example: patients prescribed a statin, minus patients with a cholesterol test on record, gives the patients on the drug with no monitoring test, a typical quality-of-care query. (Databases call this an *anti-join*.)

Watch out: a set difference is not a subtraction of sizes

With numbers, $5 - 7 = -2$ can go negative. A set difference cannot shrink below nothing: you can only cross out elements that are actually in A . So $|A \setminus B|$ is at most $|A|$, and it equals $|A| - |B|$ only in the special case $B \subseteq A$. For example, $|D \setminus H| = 3$, while $|D| - |H| = 7 - 6 = 1$.

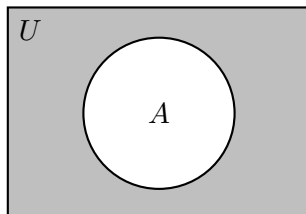
4.4 Complement

Definition: Complement

When a universal set U is fixed, the **complement** of A is everything in U outside of A :

$$\bar{A} = U \setminus A = \{x \in U \mid x \notin A\}$$

In a picture, the rectangle is the universe U , and the complement is everything in it except the circle:



Complement is the set version of **NOT**. It is simply the difference with the whole universe in front, $U \setminus A$. That is why it only makes sense once U is agreed upon: “not diabetic” means nothing until we say “among whom.” Some texts write A' instead of $\bar{A}$.

Examples.

- In the study, the non-diabetics are $\bar{D} = \{P3, P5, P7, P9, P11\}$: all twelve patients except the seven in D . As a check, $|D| + |\bar{D}| = 7 + 5 = 12 = |U|$. Every element of U is in exactly one of A and $\bar{A}$, so the two sizes always add up to $|U|$.
- Taking the complement twice gets you back where you started: the complement of $\bar{D}$ is D again. This is the set version of the double negation $\neg\neg p$, which is just p .
- The complement of the whole universe is empty, $\bar{U} = \emptyset$, and the complement of the empty set is everything, $\bar{\emptyset} = U$.

Worked example 4.2: The complement depends on the universe. Let $A = \{2, 4, 6\}$. Find $\bar{A}$ when (a) $U = \{1, 2, 3, 4, 5, 6\}$, and (b) $U = \{1, 2, \dots, 10\}$.

- (a) Everything in U outside A : $\bar{A} = \{1, 3, 5\}$. Here the complement happens to be “the odd numbers up to 6.”
- (b) Now U is bigger, and so is the complement: $\bar{A} = \{1, 3, 5, 7, 8, 9, 10\}$. It is no longer “the odd numbers”: 8 and 10 are even, but they are outside A .

The set A never changed; the universe did. Whenever you see a complement, ask “relative to what?”

Watch out: “not in A ” is not the same as “in some other named set”

Students sometimes read $\bar{D}$ as “the healthy patients.” It is not: $\bar{D}$ contains patient 7, who is hypertensive, and patient 9, who is obese. The complement of D is everyone who fails the one test “diabetic,” whatever else is true of them.

4.5 Worked Examples

Let $A = \{1, 2, 3\}$ and $B = \{2, 3, 4\}$. Here are all four of the two-set operations side by side:

$A \cup B = \{1, 2, 3, 4\}$	everything in either set
$A \cap B = \{2, 3\}$	only what is in both
$A \setminus B = \{1\}$	in A , but not in B
$B \setminus A = \{4\}$	in B , but not in A

Notice that the four answers fit together. $A \cap B$, $A \setminus B$, and $B \setminus A$ do not overlap, and together they make up $A \cup B$: $\{2, 3\}$, $\{1\}$, and $\{4\}$ combine to $\{1, 2, 3, 4\}$. Section 5 will draw these as the regions of a Venn diagram.

Procedure: evaluating a set expression

1. Work from the innermost parentheses outward, exactly as in arithmetic or in the compound statements of Unit 1-1.
2. Write each intermediate result down as a list before using it in the next step.
3. For a complement, first confirm what U is.
4. Check the final answer on one element that should be in it and one that should not, going back to the words “or,” “and,” “not.”

Worked example 4.3: All the operations on a small universe. Let $U = \{1, 2, \dots, 10\}$, let $E = \{2, 4, 6, 8, 10\}$ be the even numbers in U , and let $T = \{3, 6, 9\}$ be the multiples of 3 in U . Then:

$E \cup T = \{2, 3, 4, 6, 8, 9, 10\}$	even, or a multiple of 3 (or both)
$E \cap T = \{6\}$	even and a multiple of 3
$E \setminus T = \{2, 4, 8, 10\}$	even, but not a multiple of 3
$T \setminus E = \{3, 9\}$	a multiple of 3, but odd
$\bar{E} = \{1, 3, 5, 7, 9\}$	the odd numbers in U
$\overline{E \cup T} = \{1, 5, 7\}$	neither even nor a multiple of 3

The last line uses the procedure: first the inside, $E \cup T$, then everything in U outside it. Check with one element: 5 is odd and not a multiple of 3, so it should be in $\overline{E \cup T}$, and it is.

Worked example 4.4: A three-set expression with a complement. Keep U , E , T as above, and add $C = \{1, 2, 3, 4, 5\}$, the numbers in U that are at most 5. Find $(E \cup T) \cap \bar{C}$, $E \cap (T \cup C)$, and $\bar{E} \cap \bar{T} \cap \bar{C}$.

- $(E \cup T) \cap \bar{C}$. Inside first: $E \cup T = \{2, 3, 4, 6, 8, 9, 10\}$ from before, and $\bar{C} = \{6, 7, 8, 9, 10\}$. Keep the elements in both: $\{6, 8, 9, 10\}$. In words: numbers bigger than 5 that are even or a multiple of 3. The number 7 is bigger than 5 but is neither, so it is correctly left out.
- $E \cap (T \cup C)$. Inside first: $T \cup C = \{1, 2, 3, 4, 5, 6, 9\}$. Then keep only the even ones: $\{2, 4, 6\}$.

- $\bar{E} \cap \bar{T} \cap \bar{C}$. The three complements are $\{1, 3, 5, 7, 9\}$, $\{1, 2, 4, 5, 7, 8, 10\}$, and $\{6, 7, 8, 9, 10\}$. The only number in all three is 7, so the answer is $\{7\}$: the one number in U that is odd, not a multiple of 3, and bigger than 5.

Worked example 4.5: A clinical query on the study table. Which patients in the study are over 65 and hypertensive, but not diabetic? Let $S = \{P2, P4, P7, P10\}$ be the patients over 65. The query is

$$(S \cap H) \setminus D.$$

Inside first: $S \cap H = \{P4, P7\}$, the patients over 65 who are hypertensive (P2 and P10 are not). Then remove the diabetics: P4 is diabetic, P7 is not. The answer is $\{P7\}$. Writing the query as a set expression first, and only then computing, is a habit worth building: the expression can be checked against the English before any data is touched.

Exercise 4.1

With $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, and $C = \{3, 4, 5\}$, find:

$$A \cap C, \quad (A \cup B) \cap C, \quad A \setminus (B \cap C)$$

Solution to Exercise 4.1

Work from the inside of the parentheses outward, exactly as with compound statements in Unit 1-1.

$A \cap C$. We want the elements in both A and C . Check each element of $A = \{1, 2, 3\}$ against $C = \{3, 4, 5\}$: 1 is not in C , 2 is not in C , and 3 is. Only 3 qualifies:

$$A \cap C = \{3\}.$$

$(A \cup B) \cap C$. First the inside: $A \cup B = \{1, 2, 3, 4\}$. Then intersect with $C = \{3, 4, 5\}$: of 1, 2, 3, 4, only 3 and 4 are also in C , so

$$(A \cup B) \cap C = \{3, 4\}.$$

$A \setminus (B \cap C)$. First the inside: $B \cap C = \{3, 4\}$. Then remove those from $A = \{1, 2, 3\}$. The 3 is removed; the 4 was never in A , so there is nothing to remove. What is left is

$$A \setminus (B \cap C) = \{1, 2\}.$$

4.6 Properties of Union and Intersection

Union and intersection behave much like $+$ and $\times$ in ordinary algebra.

Commutative: *order* does not matter

$$A \cup B = B \cup A \qquad A \cap B = B \cap A$$

Associative: *grouping* does not matter

$$A \cup (B \cup C) = (A \cup B) \cup C \qquad A \cap (B \cap C) = (A \cap B) \cap C$$

Like $\wedge$ and $\vee$ in Unit 1-1, $\cup$ and $\cap$ are **binary**: each joins two sets at a time. Associativity is what lets us drop the parentheses and write $A \cup B \cup C$, exactly as for $1 + 2 + 3$.

Distributive: each distributes over the other

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

The second law looks just like $a \times (b + c) = a \times b + a \times c$, with $\cap$ playing the role of $\times$ and $\cup$ the role of $+$. The first has no arithmetic analogue: $+$ does not distribute over $\times$. For instance, $1 + (2 \times 3) = 7$, but $(1 + 2) \times (1 + 3) = 12$. With sets, both directions work.

Why should we care about these laws? Because they let you **rewrite** a query into an equivalent one that is easier to compute or easier to read, with a guarantee that the answer will not change. The laws are really the logic laws of Unit 1-1 in disguise: $\cup$ behaves like $\vee$ and $\cap$ like $\wedge$, and $\wedge$ and $\vee$ distribute over each other in exactly the same two ways.

Worked example 4.6: Checking both distributive laws on concrete sets. Take the sets of the exercise above, $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, $C = \{3, 4, 5\}$, and compute both sides of each law.

- **First law.** Left side: $B \cap C = \{3, 4\}$, so $A \cup (B \cap C) = \{1, 2, 3, 4\}$. Right side: $A \cup B = \{1, 2, 3, 4\}$ and $A \cup C = \{1, 2, 3, 4, 5\}$, so $(A \cup B) \cap (A \cup C) = \{1, 2, 3, 4\}$. The two sides agree.
- **Second law.** Left side: $B \cup C = \{2, 3, 4, 5\}$, so $A \cap (B \cup C) = \{2, 3\}$. Right side: $A \cap B = \{2, 3\}$ and $A \cap C = \{3\}$, so $(A \cap B) \cup (A \cap C) = \{2, 3\}$. Again the two sides agree.

One example does not prove a law (the laws hold for *all* sets because of the logic behind them), but checking one is an excellent way to make sure you are reading the symbols correctly.

Worked example 4.7: A distributive law in a clinical query. “Diabetic patients who are hypertensive or obese” is $D \cap (H \cup O)$. By the second law it equals $(D \cap H) \cup (D \cap O)$: “the diabetic hypertensives, together with the diabetic obese patients.” In the study, $D \cap H = \{P1, P4, P8, P12\}$ and $D \cap O = \{P2, P4, P10\}$, so the union is $\{P1, P2, P4, P8, P10, P12\}$. Computing the other way, $H \cup O = \{P1, P2, P4, P5, P7, P8, P9, P10, P12\}$, and keeping the diabetics gives the same six patients. Patient 4, who is in both $D \cap H$ and $D \cap O$, appears once.

Watch out: difference obeys none of these laws

Set difference is neither commutative nor associative. We already saw $A \setminus B \neq B \setminus A$. For grouping, with the same A, B, C :

$$(A \setminus B) \setminus C = \{1\} \setminus \{3, 4, 5\} = \{1\}, \quad A \setminus (B \setminus C) = \{1, 2, 3\} \setminus \{2\} = \{1, 3\}.$$

Different answers, so with differences the parentheses always matter, just as $(8 - 4) - 2 \neq 8 - (4 - 2)$ in arithmetic.

Watch out: never mix $\cup$ and $\cap$ without parentheses

“ $A \cup B \cap C$ ” is ambiguous, and the two readings differ:

$$(A \cup B) \cap C = \{3, 4\}, \quad A \cup (B \cap C) = \{1, 2, 3, 4\}.$$

Unlike arithmetic, where $\times$ is done before $+$ by convention, there is no agreed order between $\cup$ and $\cap$. Always write the parentheses, just as with $\wedge$ and $\vee$ in Unit 1-1.

4.7 De Morgan’s Laws for Sets

The laws from Unit 1-1 carry over word for word, with $\cup$ for OR and $\cap$ for AND.

De Morgan’s Laws (sets)

$$\begin{aligned}\overline{A \cup B} &= \bar{A} \cap \bar{B} \\ \overline{A \cap B} &= \bar{A} \cup \bar{B}\end{aligned}$$

In words: push the complement bar inward, and **the operation flips**. Union becomes intersection, and intersection becomes union.

Why they are the same laws. Follow a single element x through the second law. Being in $\overline{A \cap B}$ means *not* being in both A and B , and from there the logical De Morgan law does the work:

$$\begin{aligned}x \in \overline{A \cap B} &\Leftrightarrow \neg(x \in A \wedge x \in B) \\ &\Leftrightarrow \neg(x \in A) \vee \neg(x \in B) \quad (\text{logical De Morgan}) \\ &\Leftrightarrow x \in \bar{A} \cup \bar{B}\end{aligned}$$

The double arrow $\Leftrightarrow$ is the “if and only if” of Unit 1-1, written the way it usually is between the steps of an argument. The first law follows the same way from the other logical De Morgan law.

Set theory is logic with the statements collected into piles.

In words, the first law says: “**not in either**” is the same as “**not in the first, and not in the second.**” The second says: “**not in both**” is the same as “**missing from at least one.**”

Worked example 4.8: Checking De Morgan’s laws on a small universe. Use $U = \{1, \dots, 10\}$, $E = \{2, 4, 6, 8, 10\}$, and $T = \{3, 6, 9\}$ from before, so that $\bar{E} = \{1, 3, 5, 7, 9\}$ and $\bar{T} = \{1, 2, 4, 5, 7, 8, 10\}$.

- **First law.** Left side: $E \cup T = \{2, 3, 4, 6, 8, 9, 10\}$, so $\overline{E \cup T} = \{1, 5, 7\}$. Right side: the numbers in both $\bar{E}$ and $\bar{T}$ are 1, 5, 7. Both sides are $\{1, 5, 7\}$.
- **Second law.** Left side: $E \cap T = \{6\}$, so $\overline{E \cap T} = \{1, 2, 3, 4, 5, 7, 8, 9, 10\}$, everything except 6. Right side: $\bar{E} \cup \bar{T}$ collects every number that is odd or not a multiple of 3, which again is everything except 6. Both sides agree.

Worked example 4.9: Rewriting a patient filter. An analyst needs “the patients who are neither diabetic nor pregnant.” “Neither ... nor” means “not (diabetic or pregnant),” which is

$\overline{D \cup G}$. De Morgan's first law rewrites it as $\bar{D} \cap \bar{G}$, “not diabetic, and not pregnant.” Compute both ways in the study:

- $D \cup G = \{P1, P2, P3, P4, P6, P8, P10, P12\}$, so its complement is $\{P5, P7, P9, P11\}$.
- $\bar{D} = \{P3, P5, P7, P9, P11\}$ and $\bar{G}$ is everyone except $P3, P6, P12$. The patients in both are $\{P5, P7, P9, P11\}$.

Same four patients. In practice, the second form is often how the filter gets written in code (“diabetic is false AND pregnant is false”), and De Morgan's law is the guarantee that it means what the English says.

Watch out: the operation must flip

The most common mistake with De Morgan is to push the bar inward but forget to flip the operation, writing $\overline{E \cup T}$ as $\bar{E} \cup \bar{T}$. In the example above, $\overline{E \cup T} = \{1, 5, 7\}$ has three elements, while $\bar{E} \cup \bar{T}$ has nine. In a patient query, the wrong version turns “neither diabetic nor pregnant” into “not diabetic, or not pregnant,” which lets in almost everyone.

4.8 Back to the Cohort

Unit 1-1 opened with a criterion we could write in logic but not yet *collect*:

“Patients who are diabetic AND (hypertensive OR obese), but NOT pregnant.”

Let U be all patients in the study, and name each property as a statement about x :

$$\begin{array}{ll} d(x) = \text{“}x \text{ is diabetic”} & h(x) = \text{“}x \text{ is hypertensive”} \\ o(x) = \text{“}x \text{ is obese”} & g(x) = \text{“}x \text{ is pregnant”} \end{array}$$

In Unit 1-1 a letter such as d stood for one fixed statement. Writing $d(x)$ instead marks that the statement is about a particular patient x , so its truth value changes from patient to patient, just like “ $x \in A$.”

The cohort is a set

$$\text{Cohort} = \{ x \in U \mid d(x) \wedge (h(x) \vee o(x)) \wedge \neg g(x) \}$$

Logic supplies the rule; set-builder notation turns it into a **collection you can count**.

Each connective in the rule has a set operation to match: $\wedge$ becomes $\cap$, $\vee$ becomes $\cup$, and $\neg$ becomes a complement. So the same cohort can be written purely with the operations of this section, using the sets D, H, O, G of Section 1.3:

$$\text{Cohort} = D \cap (H \cup O) \cap \bar{G}.$$

The parentheses around $H \cup O$ are the parentheses around “hypertensive OR obese” in the English. Without them the rule would mean something else, exactly as in Unit 1-1.

Worked example 4.10: Building the cohort from the study table. Compute the cohort for our twelve patients, following the procedure for set expressions (inside out, writing down each step).

1. **Inside the parentheses.** $H \cup O$, the patients who are hypertensive or obese:

$$H \cup O = \{P1, P2, P4, P5, P7, P8, P9, P10, P12\}.$$

2. **Intersect with D .** Keep only the diabetics from that list:

$$D \cap (H \cup O) = \{P1, P2, P4, P8, P10, P12\}.$$

Patient 6 is diabetic but neither hypertensive nor obese, so the parenthesized condition already removed them.

3. **Remove the pregnant patients.** $\bar{G}$ is everyone except P3, P6, P12. Of the six patients above, only P12 is pregnant, so

$$\text{Cohort} = \{P1, P2, P4, P8, P10\}, \quad |\text{Cohort}| = 5.$$

Check one patient in and one out against the English. Patient 2 is diabetic, obese, and not pregnant: passes every part, and is in. Patient 12 is diabetic and hypertensive, but pregnant: fails the “NOT pregnant” part, and is out.

Two remarks. First, the cohort is a subset of D , as any filter that begins “diabetic AND ...” must be: $\text{Cohort} \subseteq D$. Second, the order in which we applied the three filters did not matter for the final answer (intersection is commutative and associative), which is what lets a database choose whichever order is fastest.

4.9 Extra Practice

Exercise 4.2

Let $U = \{1, 2, \dots, 12\}$, let $P = \{2, 3, 5, 7, 11\}$ be the primes in U , and let $V = \{2, 4, 6, 8, 10, 12\}$ be the even numbers in U .

1. Find $P \cap V$, $P \cup V$, $P \setminus V$, and $\overline{P \cup V}$.
2. Compute $\bar{P} \cap \bar{V}$ and compare with $\overline{P \cup V}$. Which law explains what you see?

Solution to Exercise 4.2

1. $P \cap V = \{2\}$: 2 is the only even prime. $P \cup V = \{2, 3, 4, 5, 6, 7, 8, 10, 11, 12\}$: start with V and add the odd primes 3, 5, 7, 11. $P \setminus V = \{3, 5, 7, 11\}$: the odd primes. $\overline{P \cup V} = \{1, 9\}$: the numbers in U that are neither prime nor even. (Recall that 1 is not a prime, and $9 = 3 \times 3$ is not either.)
2. $\bar{P} = \{1, 4, 6, 8, 9, 10, 12\}$ and $\bar{V} = \{1, 3, 5, 7, 9, 11\}$; the numbers in both are $\{1, 9\}$. This matches $\overline{P \cup V}$, as De Morgan’s first law promises: “not (prime or even)” is “not prime and not even.”

Exercise 4.3

Using the study table (Section 1.3), write each group as a set expression in D, H, O, G (and S , the patients over 65), then list it.

1. The hypertensive patients who are not obese.

2. The patients who are neither diabetic nor pregnant.
3. The patients who are obese or over 65.

Solution to Exercise 4.3

1. $H \setminus O$. Start from $H = \{P1, P4, P5, P7, P8, P12\}$ and cross out the obese: only P4. Result: $\{P1, P5, P7, P8, P12\}$.
2. $\overline{D \cup G}$, or equivalently $\bar{D} \cap \bar{G}$ by De Morgan. Result: $\{P5, P7, P9, P11\}$, as computed in the worked example on rewriting a filter.
3. $O \cup S$. Start from $O = \{P2, P4, P9, P10\}$ and add the members of $S = \{P2, P4, P7, P10\}$ not already present: only P7. Result: $\{P2, P4, P7, P9, P10\}$.

Exercise 4.4

With $A = \{1, 2, 3\}$ and $B = \{2, 3, 4\}$, compute $(A \setminus B) \cup (B \setminus A)$ and $(A \cup B) \setminus (A \cap B)$. What do you notice, and what does this set mean in words?

Solution to Exercise 4.4

First expression: $A \setminus B = \{1\}$ and $B \setminus A = \{4\}$, so $(A \setminus B) \cup (B \setminus A) = \{1, 4\}$. Second expression: $A \cup B = \{1, 2, 3, 4\}$ and $A \cap B = \{2, 3\}$, so $(A \cup B) \setminus (A \cap B) = \{1, 4\}$. The two are the same set. Both describe “in exactly one of the two sets”: the first as “only in A , or only in B ,” the second as “in at least one, but not in both.” This is the set version of the **exclusive OR** from Unit 1-1, and it is exactly the shape of the opening question about diabetes or hypertension “but not both,” which Section 5 answers.

Exercise 4.5

Use De Morgan’s law to rewrite “not (over 65 and diabetic)” without a NOT in front of the parentheses, and list the patients in the study it describes.

Solution to Exercise 4.5

“Over 65 and diabetic” is $S \cap D = \{P2, P4, P10\}$. Its negation is $\overline{S \cap D}$, and De Morgan’s second law gives

$$\overline{S \cap D} = \bar{S} \cup \bar{D} : \quad \text{“65 or younger, or not diabetic.”}$$

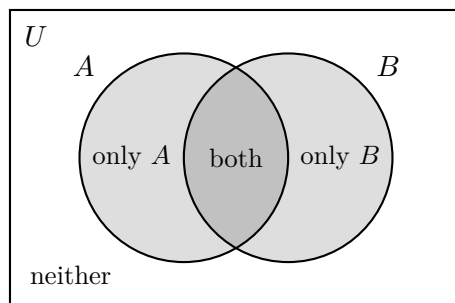
A patient qualifies by failing at least one of the two tests. That is everyone except the three patients who pass both: $\{P1, P3, P5, P6, P7, P8, P9, P11, P12\}$, nine patients. Notice that “not over 65” became “65 or younger,” not “under 65”: a patient aged exactly 65 is not *over* 65.

5 Venn Diagrams and Counting

5.1 Reading a Two-Set Venn Diagram

A **Venn diagram** draws the universe U as a rectangle and each set as a region inside it, usually a circle. Two overlapping circles cut the universe into **four** regions, and every patient sits in exactly

one of them.



Each region matches one answer to the two yes/no questions “in A ?” and “in B ?”, which is why there are $2 \times 2 = 4$ of them, just like the four rows of a two-statement truth table. Each region also has a name in the notation we have built:

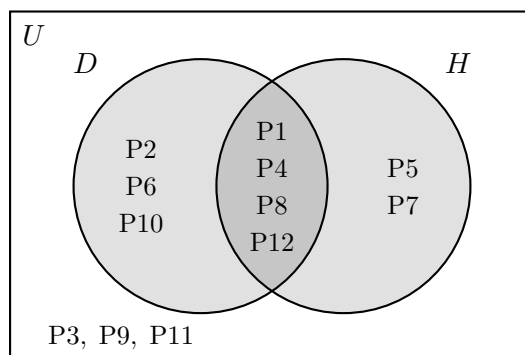
Region	In A ?	In B ?	In symbols
only A	yes	no	$A \setminus B$
both	yes	yes	$A \cap B$
only B	no	yes	$B \setminus A$
neither	no	no	$\overline{A \cup B} = \bar{A} \cap \bar{B}$

The last row has two names, and De Morgan’s law is what says they agree: “outside the union” is the same region as “outside A and outside B .”

Procedure: placing an element in a two-set Venn diagram

For each element of U , ask two yes/no questions: “in A ?” and “in B ?” The pair of answers picks out exactly one of the four regions, using the table above. Because every element gets exactly one pair of answers, every element lands in exactly one region: nobody is left out, and nobody is placed twice.

Worked example 5.1: Placing the twelve patients. Sort the study patients into the Venn diagram for D (diabetic) and H (hypertensive). Going down the table: P1 is yes/yes, so “both”; P2 is yes/no, so “only D ”; P3 is no/no, so “neither”; and so on. The finished diagram:

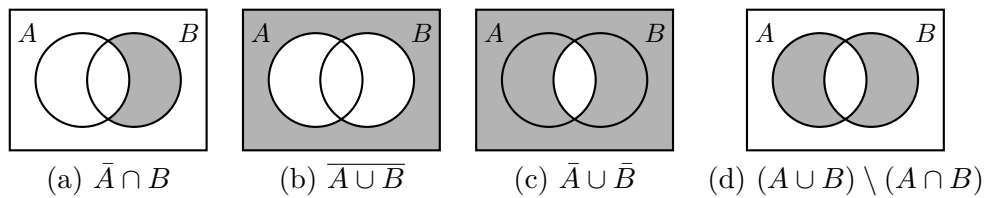


Now every question about D and H can be read straight off the picture:

- only diabetic: $D \setminus H = \{P2, P6, P10\}$, 3 patients;
- both: $D \cap H = \{P1, P4, P8, P12\}$, 4 patients;
- only hypertensive: $H \setminus D = \{P5, P7\}$, 2 patients;
- neither: $\overline{D \cup H} = \{P3, P9, P11\}$, 3 patients.

Check: $3 + 4 + 2 + 3 = 12$, the whole study. That check is worth doing every time; if the regions do not add up to $|U|$, someone was lost or double-placed.

Worked example 5.2: Shading a region for an expression. Each of the four diagrams below shades the region for one set expression. For each, decide region by region whether an element there makes the expression true.



- (a) $\bar{A} \cap B$: outside A and inside B . Only the “only B ” crescent qualifies; it is the same region as $B \setminus A$.
- (b) $\overline{A \cup B}$: outside the union, which is everything in the rectangle but outside both circles.
- (c) $\bar{A} \cup \bar{B}$: outside A or outside B . Every region qualifies except the middle lens, where an element is inside both. By De Morgan this is $\overline{A \cap B}$, and the picture shows why: it is the whole universe with the overlap cut out.
- (d) $(A \cup B) \setminus (A \cap B)$: in at least one circle, but not in both. The two crescents, without the lens. This is the “or, but not both” of the opening question.

Comparing (b) and (c) is the quickest way to see that $\overline{A \cup B}$ and $\bar{A} \cup \bar{B}$ are different sets: one is only the region outside both circles, the other is everything except the lens.

Watch out: a region can be empty

A Venn diagram always draws all four regions, even when one of them has no elements. Drawing the circles overlapping does not claim that anybody is in the overlap; if A and B are disjoint, the lens is simply an empty region, with count 0. So never infer a count from the size or shape of a region. The picture shows the *logic* of which regions exist, and the numbers written inside it carry the data.

5.2 Inclusion-Exclusion

How many elements are in $A \cup B$? Adding $|A|$ and $|B|$ counts the overlap **twice**, once as part of A and once as part of B , so we subtract it back off once:

Inclusion-exclusion (two sets)

$$|A \cup B| = |A| + |B| - |A \cap B|$$

The name describes the recipe: *include* both sets, then *exclude* the overlap that was counted twice.

Example. In a clinic of 100 patients, let D be the set of diabetics and H the set of hypertensives. (This is a larger clinic than our twelve-patient study, so here D and H are new sets with the same meaning.) Suppose $|D| = 40$, $|H| = 55$, and $|D \cap H| = 25$. Then

$$|D \cup H| = 40 + 55 - 25 = 70.$$

So 70 patients have at least one condition, and $100 - 70 = 30$ have neither. Those 30 are the “neither” region: everyone in the clinic outside $D \cup H$.

Notice that $40 + 55 = 95 \neq 70$. Forgetting to subtract the overlap is the classic error: the 25 patients with both conditions get counted twice, once among the diabetics and again among the hypertensives.

Why does subtracting $|A \cap B|$ once fix it exactly? Look at the three regions inside the circles. An element in only A is counted once by $|A|$ and not at all by $|B|$: once in total. An element in only B is likewise counted once. An element in the overlap is counted by $|A|$ *and* by $|B|$, twice in total, and the $-|A \cap B|$ takes one of those away. So every element of $A \cup B$ ends up counted exactly once. Elements outside both circles are never counted at all, as they should not be.

Most two-set counting problems ask for one of five things, and each has a formula that follows from the regions:

In words	As a set	How to count it
at least one (“ A or B ”)	$A \cup B$	$ A + B - A \cap B $
both (“ A and B ”)	$A \cap B$	usually given, or solved for
only A (“ A but not B ”)	$A \setminus B$	$ A - A \cap B $
exactly one (“but not both”)	$(A \cup B) \setminus (A \cap B)$	$ A \cup B - A \cap B $
neither	$\overline{A \cup B}$	$ U - A \cup B $

Procedure: a two-set counting problem

1. Name the universe and the two sets, and write down every number you are given in set notation.
2. Find the overlap $|A \cap B|$ first, if it is not given (see the second worked example below).
3. Fill in the Venn diagram from the middle outward: the overlap, then the two “only” regions, then “neither.”
4. Answer the question from the regions, and check that the four regions add up to $|U|$.

Worked example 5.3: Inclusion-exclusion on the study table. In our study, $|D| = 7$, $|H| = 6$, and $|D \cap H| = 4$. The formula gives

$$|D \cup H| = 7 + 6 - 4 = 9,$$

which matches the direct count from the placed diagram in the previous subsection: $3 + 4 + 2 = 9$ patients inside the circles. With twelve patients we could count directly; the formula is what we use when the table has twelve thousand rows and all we have are the three totals.

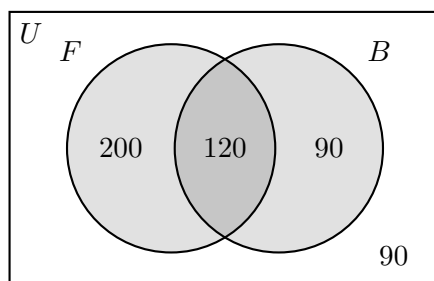
Worked example 5.4: Solving for the overlap. A hospital has 500 staff. A survey finds that 320 received the flu vaccine, 210 received a COVID booster, and 90 received neither. How many received both? Let F be the flu-vaccinated staff and B the boosted staff. The overlap is the unknown, so use the other facts to get $|F \cup B|$ first. Everyone outside “neither” got at least one shot:

$$|F \cup B| = 500 - 90 = 410.$$

Now inclusion-exclusion has only one unknown:

$$410 = 320 + 210 - |F \cap B| \implies |F \cap B| = 530 - 410 = 120.$$

So 120 staff received both. Fill in the rest from the middle outward: flu only is $320 - 120 = 200$, booster only is $210 - 120 = 90$, and neither is 90.



Check: $200 + 120 + 90 + 90 = 500$. (Two regions happen to hold 90 each, booster only and neither; that is a coincidence of the numbers, not a rule.)

Worked example 5.5: Merging two patient lists. An analyst pulls two lists from an EHR: 1,240 patients had an HbA1c test (a diabetes blood test) this year, and 860 had a lipid panel. Matching IDs shows 510 patients had both. How many distinct patients had at least one of the two tests?

$$|A \cup L| = 1240 + 860 - 510 = 1590.$$

If the analyst had simply stacked the two lists without removing duplicates, the combined file would have $1,240 + 860 = 2,100$ rows, but only 1,590 distinct patients; the 510 patients on both lists would each appear twice.

Worked example 5.6: Disjoint sets need no correction. In our study, the pregnant patients and the male patients do not overlap: $G \cap M = \emptyset$. So

$$|G \cup M| = |G| + |M| - 0 = 3 + 5 = 8.$$

For disjoint sets, and only for disjoint sets, the size of the union is the plain sum of the sizes. Directly: $G \cup M = \{P2, P3, P4, P6, P7, P8, P11, P12\}$, eight patients.

Worked example 5.7: Catching impossible data. A report on 100 patients states that 60 smoke, 70 drink alcohol, and 20 do both. Is that possible? Inclusion-exclusion gives

$$|S \cup A| = 60 + 70 - 20 = 110.$$

But $S \cup A$ is a subset of the 100 patients, so it cannot have 110 elements. The report must contain an error. In fact the overlap must be at least 30: we need $60 + 70 - |S \cap A| \leq 100$, that is, $|S \cap A| \geq 130 - 100 = 30$. When two large groups share a small universe, they are *forced* to overlap. Sanity checks like this one catch data-entry and coding errors before they reach a published table.

Watch out: the three sanity checks

Whatever the problem, the counts must satisfy

- the overlap is no bigger than the smaller set: $|A \cap B| \leq |A|$ and $|A \cap B| \leq |B|$;
- the union is at least as big as the bigger set, and no bigger than the universe;
- the four regions are never negative, and they add up to $|U|$.

A negative region, such as “−5 patients in only A ,” always means a number was misread or the overlap was subtracted from the wrong place.

5.3 Answering the Opening Question

We can now answer the question from Section 1:

*“How many patients have diabetes **or** hypertension, but **not both**?”*

In symbols, that is the region $(D \cup H) \setminus (D \cap H)$: the two crescents, but not the middle. The middle, $D \cap H$, sits entirely inside $D \cup H$, so removing it takes away exactly $|D \cap H|$ patients:

$$|(D \cup H) \setminus (D \cap H)| = |D \cup H| - |D \cap H|.$$

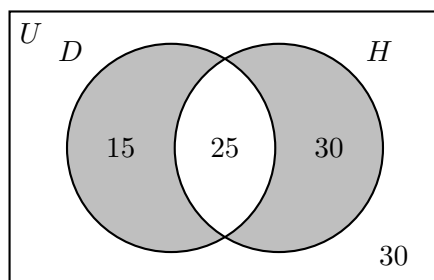
With the numbers from the clinic example:

$$70 - 25 = 45.$$

As a check, count the two crescents directly. Diabetes only is $40 - 25 = 15$, hypertension only is $55 - 25 = 30$, and $15 + 30 = 45$.

Recognize the shape: “or, but not both” is the **exclusive OR** from Unit 1-1. It is the same idea, now counted.

Here is the whole clinic in one picture, filled in from the middle outward. The two shaded crescents are the answer.



There is a third way to get the same number, straight from the three given totals. The “exactly one” count is $|D| + |H|$ with the overlap removed from *both* terms, since the overlap sits inside D and inside H :

$$(|D| - |D \cap H|) + (|H| - |D \cap H|) = |D| + |H| - 2|D \cap H| = 40 + 55 - 50 = 45.$$

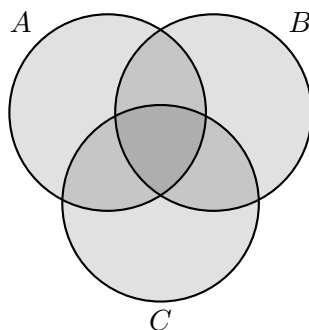
Compare with inclusion-exclusion: for “at least one” we subtract the overlap once (it was counted twice, and should count once); for “exactly one” we subtract it twice (it was counted twice, and should not count at all).

Worked example 5.8: Exactly one, in the exercise survey. Use the survey in the exercise at the end of this section: 120 patients, $|E| = 70$ exercise, $|D| = 45$ diet, $|E \cap D| = 25$ both. How many do exactly one of the two? By the formula, $70 + 45 - 2 \times 25 = 65$. By regions: exercise only is $70 - 25 = 45$, diet only is $45 - 25 = 20$, and $45 + 20 = 65$. Both routes agree.

Worked example 5.9: Exactly one, in the study table. Which study patients are diabetic or hypertensive, but not both? From the placed diagram, the two crescents hold $\{P2, P6, P10\}$ and $\{P5, P7\}$, so the answer is $\{P2, P5, P6, P7, P10\}$, five patients. The formula agrees: $|D| + |H| - 2|D \cap H| = 7 + 6 - 8 = 5$.

5.4 Three-Set Venn Diagrams

Three circles cut the universe into **eight** regions: 2^3 , one for each in/out combination, exactly like the eight rows of a three-statement truth table.



The shading darkens where circles overlap, so the center, which lies in all three sets, is the darkest region.

Strategy: fill in from the middle outward

Start with $|A \cap B \cap C|$, then the two-way overlaps (subtracting the center), then the single regions. Each region gets counted exactly once.

The order matters because the center is part of every overlap. For example, the patients in A and B but not C number $|A \cap B| - |A \cap B \cap C|$, which you can only compute once the center is known. In the same way, each single region is its whole circle minus the overlap regions already filled in.

Written out step by step, with $t = |A \cap B \cap C|$ for the center:

Procedure: filling a three-set Venn diagram from the middle outward

1. **Center:** $t = |A \cap B \cap C|$.
2. **Two-way regions** (in exactly two sets): $|A \cap B| - t$, $|A \cap C| - t$, $|B \cap C| - t$.
3. **One-way regions** (in exactly one set): each whole circle minus its two two-way regions and the center. For A :

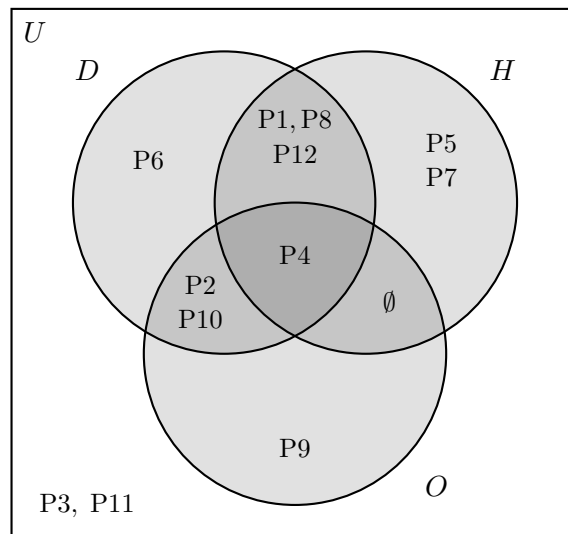
$$\text{only } A = |A| - (|A \cap B| - t) - (|A \cap C| - t) - t.$$

4. **Outside:** $|U|$ minus the sum of the seven regions inside the circles.
5. **Check:** no region is negative, and all eight add up to $|U|$.

Each region corresponds to one row of a three-statement truth table: “in A ?”, “in B ?”, “in C ?” each answered yes or no.

Worked example 5.10: The study table in three sets. Sort the twelve study patients by diabetic (D), hypertensive (H), and obese (O). For each patient, read the three answers off the table:

In D ?	In H ?	In O ?	Region	Patients
yes	yes	yes	center	P4
yes	yes	no	D and H only	P1, P8, P12
yes	no	yes	D and O only	P2, P10
yes	no	no	only D	P6
no	yes	yes	H and O only	(none)
no	yes	no	only H	P5, P7
no	no	yes	only O	P9
no	no	no	outside	P3, P11



Check: $1 + 3 + 2 + 1 + 0 + 2 + 1 + 2 = 12$. Notice the empty region, marked $\emptyset$: nobody in the study is hypertensive and obese without also being diabetic. The diagram still draws that region; its count is simply 0.

Now run the procedure the other way, as if we only had totals, to see it recover the same numbers. The totals are $|D| = 7$, $|H| = 6$, $|O| = 4$, $|D \cap H| = 4$, $|D \cap O| = 3$, $|H \cap O| = 1$, and $|D \cap H \cap O| = 1$.

1. Center: $t = 1$.
2. Two-way: D and H only $= 4 - 1 = 3$; D and O only $= 3 - 1 = 2$; H and O only $= 1 - 1 = 0$.
3. One-way: only $D = 7 - 3 - 2 - 1 = 1$; only $H = 6 - 3 - 0 - 1 = 2$; only $O = 4 - 2 - 0 - 1 = 1$.
4. Outside: $12 - (1 + 3 + 2 + 0 + 1 + 2 + 1) = 12 - 10 = 2$.

Every count matches the table. The procedure is how you fill in a diagram when you have only the totals, which is the usual situation with large data.

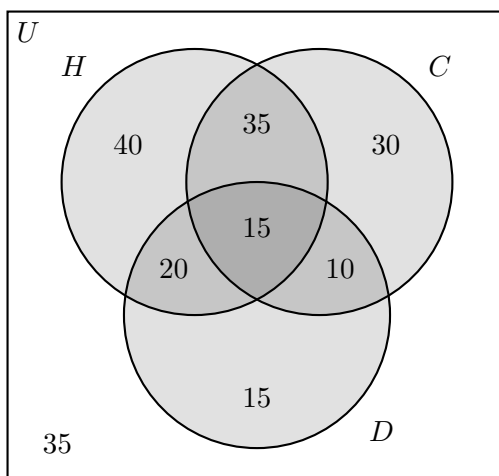
Worked example 5.11: A cardiology survey. A cardiology clinic surveys 200 patients about three conditions: hypertension (H), high cholesterol (C), and diabetes (D). The results are

$$\begin{aligned} |H| &= 110, & |C| &= 90, & |D| &= 60, \\ |H \cap C| &= 50, & |H \cap D| &= 35, & |C \cap D| &= 25, & |H \cap C \cap D| &= 15. \end{aligned}$$

How many patients have none of the three? Exactly one? Exactly two?

Fill in from the middle outward.

1. Center: 15.
2. Two-way: H and C only $= 50 - 15 = 35$; H and D only $= 35 - 15 = 20$; C and D only $= 25 - 15 = 10$.
3. One-way: only $H = 110 - 35 - 20 - 15 = 40$; only $C = 90 - 35 - 10 - 15 = 30$; only $D = 60 - 20 - 10 - 15 = 15$.
4. Inside the circles: $15 + 35 + 20 + 10 + 40 + 30 + 15 = 165$. Outside: $200 - 165 = 35$.



Now the questions read off the diagram. **None:** 35 patients. **Exactly one:** $40 + 30 + 15 = 85$. **Exactly two:** $35 + 20 + 10 = 65$. **All three:** 15. Check: $35 + 85 + 65 + 15 = 200$, since every patient has none, exactly one, exactly two, or all three conditions.

Worked example 5.12: When the center is the unknown. A registry of 250 patients records three risk factors: smoking (S), high cholesterol (C), and a family history of heart disease (F). We know

$$\begin{aligned} |S| &= 90, & |C| &= 120, & |F| &= 100, \\ |S \cap C| &= 40, & |S \cap F| &= 30, & |C \cap F| &= 50, \end{aligned}$$

and that 45 patients have none of the three. How many have all three?

The center is unknown, so call it t and fill in the diagram in terms of t , exactly as the procedure says.

- Two-way regions: $40 - t$, $30 - t$, $50 - t$.
- One-way regions:

$$\begin{aligned} \text{only } S &= 90 - (40 - t) - (30 - t) - t = 20 + t, \\ \text{only } C &= 120 - (40 - t) - (50 - t) - t = 30 + t, \\ \text{only } F &= 100 - (30 - t) - (50 - t) - t = 20 + t. \end{aligned}$$

The seven inside regions must hold everyone except the 45 with no risk factor, so they add up to $250 - 45 = 205$:

$$t + (40 - t) + (30 - t) + (50 - t) + (20 + t) + (30 + t) + (20 + t) = 205.$$

The t 's nearly cancel: four $+t$ and three $-t$ leave a single t , and the numbers add to 190. So $190 + t = 205$, and $t = 15$. Fifteen patients have all three risk factors. Substituting back, the regions are 15 (center), 25, 15, 35 (two-way), and 35, 45, 35 (one-way); they add to 205, and none is negative, so the answer is consistent.

Watch out: a pairwise total is not a “two-way only” region

The number $|A \cap B|$ counts *everyone* in both A and B , including the people in the center who are also in C . Writing $|A \cap B|$ directly into the “ A and B only” region counts the center patients twice. That is why the center goes in first, and why it is subtracted from every pairwise total before anything else is filled in.

The exercise below puts the two-set tools from this section together.

Exercise 5.1

A clinic surveys 120 patients. Let E be the set who exercise regularly, and D the set who follow a special diet:

$$|E| = 70, \quad |D| = 45, \quad |E \cap D| = 25$$

1. How many do **at least one** of the two?
2. How many do **neither**?
3. How many exercise but do **not** diet?

Solution to Exercise 5.1

1. **At least one: inclusion-exclusion.** “At least one” is the union $E \cup D$:

$$|E \cup D| = |E| + |D| - |E \cap D| = 70 + 45 - 25 = 90.$$

2. **Neither: everything outside that union.** The universe is the 120 patients surveyed, so

$$|\overline{E \cup D}| = 120 - 90 = 30.$$

3. **Exercise but not diet: a difference.**

$$|E \setminus D| = |E| - |E \cap D| = 70 - 25 = 45.$$

The 25 who do both are already counted inside E , so subtract them off.

Check. The four regions of the diagram are exercise only (45), both (25), diet only ($45 - 25 = 20$), and neither (30). Every patient sits in exactly one region, so these must add up to the whole survey, and they do: $45 + 25 + 20 + 30 = 120$.

5.5 Extra Practice

Exercise 5.2

Of 300 students, 180 drink coffee, 120 drink tea, and 60 drink both. How many drink at least one? Neither? Coffee but not tea? Exactly one of the two?

Solution to Exercise 5.2

Let C be the coffee drinkers and T the tea drinkers, with $|C \cap T| = 60$ in the middle.

- At least one: $|C \cup T| = 180 + 120 - 60 = 240$.
- Neither: $300 - 240 = 60$.
- Coffee but not tea: $|C \setminus T| = 180 - 60 = 120$.
- Exactly one: tea only is $120 - 60 = 60$, so exactly one is $120 + 60 = 180$. (Or: $|C \cup T| - |C \cap T| = 240 - 60 = 180$.)

Check the four regions: $120 + 60 + 60 + 60 = 300$.

Exercise 5.3

In a clinic, 90 patients take at least one of two medications, A or B . 50 take A , and 15 take both. How many take B ?

Solution to Exercise 5.3

Inclusion-exclusion links all four numbers, and only $|B|$ is unknown:

$$|A \cup B| = |A| + |B| - |A \cap B| \implies 90 = 50 + |B| - 15 \implies |B| = 90 - 50 + 15 = 55.$$

Check with regions: A only is $50 - 15 = 35$, B only is $55 - 15 = 40$, and $35 + 15 + 40 = 90$ patients take at least one. Notice the sign: the overlap is *added* back when solving for $|B|$, because it was subtracted in the formula.

Exercise 5.4

A pharmacy reviews 80 patients for three drug classes: A (statins), B (beta blockers), C (ACE inhibitors).

$$\begin{aligned} |A| &= 38, & |B| &= 30, & |C| &= 25, \\ |A \cap B| &= 12, & |A \cap C| &= 9, & |B \cap C| &= 7, & |A \cap B \cap C| &= 4. \end{aligned}$$

Fill in all eight regions of the Venn diagram. How many patients take none of the three? Exactly one? At least two?

Solution to Exercise 5.4

From the middle outward:

- Center: 4.
- Two-way: A and B only $= 12 - 4 = 8$; A and C only $= 9 - 4 = 5$; B and C only $= 7 - 4 = 3$.
- One-way: only $A = 38 - 8 - 5 - 4 = 21$; only $B = 30 - 8 - 3 - 4 = 15$; only $C = 25 - 5 - 3 - 4 = 13$.
- Inside: $4 + 8 + 5 + 3 + 21 + 15 + 13 = 69$. Outside: $80 - 69 = 11$.

So 11 take none of the three, $21 + 15 + 13 = 49$ take exactly one, and $8 + 5 + 3 + 4 = 20$ take at least two (exactly two, or all three). Check: $11 + 49 + 20 = 80$. For a pharmacist, “at least two” is the interesting group: those are the patients to screen for drug interactions.

Exercise 5.5

In a class of 60 students, 30 take statistics (S), 25 take biology (B), and 20 take chemistry (C); 10 take S and B , 8 take S and C , 6 take B and C , and 3 take all three. How many take none of the three, and how many take exactly one?

Solution to Exercise 5.5

Center 3. Two-way only: $10 - 3 = 7$, $8 - 3 = 5$, $6 - 3 = 3$. One-way: only $S = 30 - 7 - 5 - 3 = 15$, only $B = 25 - 7 - 3 - 3 = 12$, only $C = 20 - 5 - 3 - 3 = 9$. Inside the circles: $3 + 7 + 5 + 3 + 15 + 12 + 9 = 54$, so $60 - 54 = 6$ students take none. Exactly one: $15 + 12 + 9 = 36$.

6 Sets of Numbers

6.1 The Standard Number Systems

A handful of sets of numbers are used so often that each gets its own “blackboard bold” letter. The “...” in a list means the pattern continues forever.

- **Natural numbers:** $\mathbb{N} = \{1, 2, 3, 4, \dots\}$, the counting numbers.
- **Whole numbers:** $\mathbb{W} = \{0, 1, 2, 3, 4, \dots\}$, the naturals, plus zero.
- **Integers:** $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$, the wholes, plus negatives.
- **Rational numbers:** $\mathbb{Q} = \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0 \right\}$, ratios of integers. Read it as “all fractions p/q such that p and q are integers and q is not zero”; the $q \neq 0$ is there because dividing by zero is undefined.
- **Real numbers:** $\mathbb{R}$, every rational **and** every irrational number, that is, every point on the number line. The **irrational** numbers, such as $\sqrt{2}$ and π , are the ones that cannot be written as a ratio of integers; we look at them more closely below.
- **Complex numbers:** $\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\}$, a real part plus an imaginary part, where $i = \sqrt{-1}$. The number i is defined by $i^2 = -1$, something no real number can do.

Some books start $\mathbb{N}$ at 0. In this course $\mathbb{N}$ starts at 1, and $\mathbb{W}$ is the version that includes 0.

Each one sits inside the next

$$\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$$

Check each link. Every natural number is a whole number; every whole number is an integer; every integer n is rational, since $n = \frac{n}{1}$; every rational is real; and every real number a is complex, since $a = a + 0i$. Each containment is *proper*, because each step adds something new: zero, then the negatives, then the fractions, then the irrationals, then the imaginary part.

The imaginary numbers are **not** a link in that chain: no real number squares to a negative, so $2i \notin \mathbb{R}$. They sit inside $\mathbb{C}$ alongside $\mathbb{R}$, not around it. (The letter $\mathbb{Z}$ comes from the German *Zahlen*, “numbers”; $\mathbb{Q}$ is for *quotient*.)

Why so many systems? Each answers a different kind of question.

- $\mathbb{N}$ answers “how many?” when the answer cannot be zero: the number of beds in a ward that exists, the position of a patient in a queue.
- $\mathbb{W}$ answers “how many?” when zero is possible: missed doses this week, ER visits last year.
- $\mathbb{Z}$ allows a *direction*: a change in the number of occupied beds since yesterday (+3 or −2), a temperature below zero.
- $\mathbb{Q}$ allows *parts*: a patient took 27 of 30 prescribed doses, an adherence of $\frac{27}{30} = 0.9$.
- $\mathbb{R}$ allows *every point on a continuous scale*: an idealized exact weight, blood pressure, or time.
- $\mathbb{C}$ is needed later in mathematics (for example, some equations have no real solutions but do have complex ones); in this course it mostly appears as the outer box of the picture.

Data scientists meet the same distinctions as **variable types**. A *count* variable lives in $\mathbb{W}$; a *continuous* measurement is modeled as living in $\mathbb{R}$. Knowing which set a variable lives in tells you which values are even possible, and so which values in a dataset must be errors: an ER-visit count of 2.5, or of -1 , is not a value that $\mathbb{W}$ contains.

Examples and non-examples for each set.

Set	Some members	Some non-members
$\mathbb{N}$	1, 12, 1000, $\frac{8}{2}$ (which is 4)	0, -1 , $\frac{1}{2}$
$\mathbb{W}$	0, 1, 12	-1 , $\frac{1}{2}$, 0.3
$\mathbb{Z}$	-7 , 0, 5, $-\frac{12}{4}$ (which is -3)	$\frac{1}{2}$, -0.5 , $\sqrt{2}$
$\mathbb{Q}$	$\frac{3}{4}$, $-0.5 = -\frac{1}{2}$, $7 = \frac{7}{1}$, $0.333\dots = \frac{1}{3}$	$\sqrt{2}$, π
$\mathbb{R}$	$\sqrt{2}$, π , -4.25 , 0	i , $2i$, $3 + 2i$

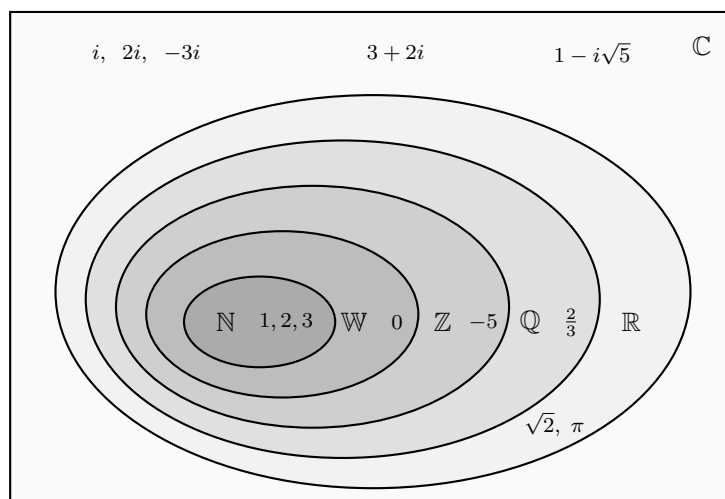
Notice the non-members in the last row: to fail to be real, a number has to leave the number line altogether.

Watch out: judge the number, not how it is written

Whether a number belongs to a set depends on its *value*, not its appearance. $\frac{8}{2}$ is written as a fraction, but its value is 4, a natural number. 4.0 has a decimal point, but it is also 4. $\sqrt{9}$ has a root sign, but it is 3. Always simplify first, then classify.

6.2 The Number Systems, Nested

The same chain as a picture. Each number is placed in the *smallest* set it belongs to, and it also belongs to every set drawn around that one.



Reading outward: 1, 2, 3 are natural; 0 is whole but not natural; -5 is an integer but not whole; $\frac{2}{3}$ is rational but not an integer. Inside $\mathbb{R}$ but outside $\mathbb{Q}$ are the **irrational**s, such as $\sqrt{2}$ and π , which are not ratios of integers. Inside $\mathbb{C}$ but outside $\mathbb{R}$ is everything with a nonzero **imaginary** part: i is nowhere on the real line.

The picture turns classifying a number into a two-step job.

Procedure: classifying a number

1. **Simplify** it to its plainest form: compute square roots that come out exactly, reduce fractions, drop trailing zeros.
2. **Find the smallest set** in the chain $\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$ that contains it, asking in order: a positive whole number? zero? a negative whole number? a ratio of integers? a point on the number line?
3. The number then belongs to that set **and every set to its right** in the chain, and to none to its left.

6.3 Rational vs. Irrational

How can you tell a rational number from an irrational one? Look at its decimal expansion.

A rational number's decimal expansion either stops or repeats

- $\frac{1}{2} = 0.5$ (terminates)
- $\frac{1}{3} = 0.3333\dots$ (repeats)
- $\frac{1}{7} = 0.142857\,142857\dots$ (repeats, with period 6)

“Period 6” means the repeating block, 142857, is six digits long, and that block repeats forever.

An irrational number's decimal expansion does neither

- $\sqrt{2} = 1.41421356\dots$
- $\pi = 3.14159265\dots$
- $e = 2.71828182\dots$

Here the “...” means the digits go on forever *without* ever settling into a repeating block. (e is another famous constant, which we will meet again later in the course.)

This has a practical consequence that Unit 1-5 will pay off. A computer stores only finitely many digits, so it never holds π , or even $\frac{1}{3}$, exactly. That is the subject of **floating-point arithmetic**.

Why must a fraction's decimal stop or repeat? Watch long division at work on $\frac{1}{7}$. At each step we bring down a 0, divide by 7, write the digit, and carry the *remainder* to the next step:

Step	Divide	Digit	Remainder
1	$10 \div 7$	1	3
2	$30 \div 7$	4	2
3	$20 \div 7$	2	6
4	$60 \div 7$	8	4
5	$40 \div 7$	5	5
6	$50 \div 7$	7	1
7	$10 \div 7$	1	3

After step 6 the remainder is 1, exactly the situation we started in (we began by dividing 1 by 7). So step 7 is a copy of step 1, step 8 will be a copy of step 2, and so on: the digits 142857 come around again, forever. And it *had* to happen. Dividing by 7, the only possible remainders are 0, 1, 2, 3, 4, 5, 6. If a remainder is ever 0, the division stops and the decimal terminates. If not, there are only six other remainders available, so within at most six steps some remainder must come back, and the digits cycle from there. The same argument works for any fraction $\frac{p}{q}$: at most q possible remainders, so the decimal stops or repeats within q steps.

Two more to compare:

- $\frac{3}{8}$: the remainders are 6, 4, 0. The remainder 0 ends the division, giving $\frac{3}{8} = 0.375$, which terminates.
- $\frac{5}{6}$: the first step ($50 \div 6$) gives digit 8, remainder 2; the next ($20 \div 6$) gives digit 3, remainder 2 again. So $\frac{5}{6} = 0.8333\dots$: one digit that does not repeat, then a repeating 3. “Repeats” allows a few non-repeating digits at the start.

An irrational number has no such mechanism behind it: it is not a division of integers, so nothing forces its digits to cycle, and they never do.

Worked example 6.1: Rational or irrational, from the decimal. Decide which of these are rational.

1. $0.363636\dots$, with the block 36 repeating forever. It repeats, so it is **rational**. In fact it is $\frac{4}{11}$: long division of 4 by 11 gives digits 3, 6, 3, 6, $\dots$
2. 0.625. It terminates, so it is **rational**: $0.625 = \frac{625}{1000} = \frac{5}{8}$.
3. $0.1010010001\dots$, where each run of zeros is one longer than the last. There is a clear *pattern*, but no fixed block of digits repeats forever, since the runs of zeros keep growing. So it is **irrational**.
4. $0.58333\dots$ (that is, 0.58 followed by 3’s forever). It eventually repeats, so it is **rational**; it equals $\frac{7}{12}$.

Watch out: $\frac{22}{7}$ is not π

School often uses $\frac{22}{7}$ for π . It is only an approximation:

$$\frac{22}{7} = 3.142857142857\dots \quad \text{while} \quad \pi = 3.14159265\dots$$

They agree only to two decimal places. $\frac{22}{7}$ is rational (its decimal repeats with period 6, like $\frac{1}{7}$),

and π is irrational. Likewise, the 3.14159265 on a calculator screen is a terminating decimal, hence rational: a rational stand-in for an irrational number.

A data-science consequence. Every number actually stored in a data table has finitely many digits, so every stored value is rational. A weight recorded as 72.4 kg is $\frac{724}{10}$. The real numbers are the mathematical ideal behind a measurement, and the stored data are rational approximations of it. In Python, $1/3$ is displayed as 0.3333333333333333, sixteen 3's and then it stops, and $0.1 + 0.2$ comes out as 0.30000000000000004 rather than 0.3. Unit 1-5 explains exactly why.

Worked example 6.2: Classifying eight numbers. Place each number in every one of $\mathbb{N}, \mathbb{W}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ that contains it, using the procedure: simplify, find the smallest set, then take everything to its right.

1. $-\frac{12}{4}$. Simplify: $-\frac{12}{4} = -3$. A negative whole number, so its smallest set is $\mathbb{Z}$: it is in $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$.
2. 0.75. Simplify: $0.75 = \frac{3}{4}$, a ratio of integers that is not a whole number. Smallest set $\mathbb{Q}$: in $\mathbb{Q}, \mathbb{R}, \mathbb{C}$.
3. $\sqrt{16}$. Simplify: $\sqrt{16} = 4$. Smallest set $\mathbb{N}$: in all six.
4. $\frac{0}{5}$. Simplify: $\frac{0}{5} = 0$ (zero divided by anything nonzero is zero). Smallest set $\mathbb{W}$: in $\mathbb{W}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$.
5. $\frac{22}{7}$. Already in lowest terms, and not a whole number. Smallest set $\mathbb{Q}$: in $\mathbb{Q}, \mathbb{R}, \mathbb{C}$. (Close to π , but rational.)
6. π . Irrational, but a point on the number line. Smallest set $\mathbb{R}$: in $\mathbb{R}, \mathbb{C}$.
7. 0.1010010001... (growing runs of zeros). Its decimal neither stops nor repeats, so it is irrational. Smallest set $\mathbb{R}$: in $\mathbb{R}, \mathbb{C}$.
8. $\sqrt{-4}$. No real number squares to -4 , so this is not on the number line. It is $2i$, since $(2i)^2 = 4i^2 = -4$. Smallest set $\mathbb{C}$: in $\mathbb{C}$ only.

	$\mathbb{N}$	$\mathbb{W}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	$\mathbb{C}$
$-\frac{12}{4}$			✓	✓	✓	✓
0.75				✓	✓	✓
$\sqrt{16}$	✓	✓	✓	✓	✓	✓
$\frac{0}{5}$		✓	✓	✓	✓	✓
$\frac{22}{7}$				✓	✓	✓
π					✓	✓
0.1010010001...					✓	✓
$\sqrt{-4}$						✓

Every row is a run of checks with no gaps, starting at the smallest set and going all the way right. A row with a gap (say, checked under $\mathbb{Z}$ but not under $\mathbb{Q}$) would contradict the nesting, and is a sign of a mistake.

Worked example 6.3: Which set does a variable live in? For each quantity from a health dataset, name the smallest of $\mathbb{N}, \mathbb{W}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ that holds *every possible value*.

1. **Number of hospital admissions last year.** Could be $0, 1, 2, \dots$, never negative and never fractional: $\mathbb{W}$.
2. **Number of people in a patient's household, counting the patient.** At least 1, always whole: $\mathbb{N}$.
3. **Change in the number of occupied ICU beds since yesterday.** Can be positive, zero, or negative, but always whole: $\mathbb{Z}$.
4. **Fraction of prescribed doses taken, out of 30.** Values like $\frac{27}{30}$ or $\frac{30}{30}$: always a ratio of integers, so $\mathbb{Q}$.
5. **The exact length of the diagonal of a square tile of side 1 cm.** By the Pythagorean theorem it is $\sqrt{2}$ cm, which is irrational: $\mathbb{R}$. Continuous measurements in general are modeled in $\mathbb{R}$.

Choosing the right set is the first step of any data check. An admissions count of 1.5 is not in $\mathbb{W}$, and a household size of 0 is not in $\mathbb{N}$, so both must be errors. (Some checks need more than the number system: an adherence of $\frac{32}{30}$ is a perfectly good rational number, but it claims more doses were taken than were prescribed.)

Exercise 6.1

For each number below, name **every** one of $\mathbb{N}, \mathbb{W}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ that it belongs to: put a check in each box that applies.

	$\mathbb{N}$	$\mathbb{W}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$
7					
-3					
0					
$\frac{5}{8}$					
$\sqrt{9}$					
$\sqrt{5}$					

Solution to Exercise 6.1

Because the sets are nested, once a number is in one set it is in every set to the right of it as well. So for each number, find the *smallest* set it belongs to and check everything from there on.

	$\mathbb{N}$	$\mathbb{W}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	Why
7	✓	✓	✓	✓	✓	a counting number
-3			✓	✓	✓	negative
0		✓	✓	✓	✓	$\mathbb{N}$ starts at 1
$\frac{5}{8}$				✓	✓	a ratio of integers
$\sqrt{9}$	✓	✓	✓	✓	✓	$\sqrt{9} = 3$
$\sqrt{5}$					✓	irrational

The two traps. First, $\sqrt{9}$ is simply 3: a square root is not automatically irrational. It is irrational only when no ratio of integers squares to the number under the root, as with $\sqrt{5}$. Second, 0 lands in $\mathbb{W}$ but not $\mathbb{N}$, since we take $\mathbb{N} = \{1, 2, 3, \dots\}$.

6.4 Implications Between Number Systems

Recall from Section 3 that $A \subseteq B$ means “ $x \in A \rightarrow x \in B$ ” for every x . So the chain of number systems can also be read as a chain of implications, and the logic of Unit 1-1 applies to it directly.

Exercise 6.2

True or false? If false, give a counterexample.

1. $x \in \mathbb{N} \rightarrow x \in \mathbb{Z}$
2. $x \in \mathbb{R} \rightarrow x \in \mathbb{Z}$
3. $x \in \mathbb{Z} \rightarrow x \in \mathbb{Q}$
4. $x \in \mathbb{Q} \rightarrow x \in \mathbb{R}$
5. $x \notin \mathbb{Q} \rightarrow x \notin \mathbb{R}$
6. $x \notin \mathbb{Z} \rightarrow x \notin \mathbb{N}$

Recall from Unit 1-1 that an implication fails only when the left side is **true** and the right side is **false**, so one counterexample settles it.

Solution to Exercise 6.2

Each claim is meant for *every* x . To show one is false, we need a single x that makes the left side true and the right side false.

Claim	T/F	Why
1. $x \in \mathbb{N} \rightarrow x \in \mathbb{Z}$	T	every natural number is an integer
2. $x \in \mathbb{R} \rightarrow x \in \mathbb{Z}$	F	$x = 0.5$ is real but not an integer
3. $x \in \mathbb{Z} \rightarrow x \in \mathbb{Q}$	T	$n = \frac{n}{1}$, a ratio of integers
4. $x \in \mathbb{Q} \rightarrow x \in \mathbb{R}$	T	every rational sits on the real line
5. $x \notin \mathbb{Q} \rightarrow x \notin \mathbb{R}$	F	$x = \sqrt{2}$ is irrational and real
6. $x \notin \mathbb{Z} \rightarrow x \notin \mathbb{N}$	T	the contrapositive of #1

For #2, $x = 0.5$ makes the left side true (0.5 is real) and the right side false (0.5 is not an integer). For #5, $x = \sqrt{2}$ makes the left side true ($\sqrt{2}$ is not rational) and the right side false ($\sqrt{2}$ is real). For #6: if x is not even an integer, it certainly cannot be a natural number. This is the contrapositive of #1, and Unit 1-1 showed that a contrapositive is always equivalent to the original, so it is true because #1 is.

Notice what these really ask. “ $x \in A \rightarrow x \in B$ ” is true for **every** x exactly when $A \subseteq B$, so the true rows are precisely the containments in $\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$, read left to right (#6 is #1 in contrapositive form). Reading a containment *right to left*, as #2 does, breaks it: that is the converse. So does negating both sides without flipping the order, as #5 does to #4: that is the inverse. Neither the converse nor the inverse is equivalent to the original.

Worked example 6.4: Three more implications. Decide each, with a reason or a counterexample.

1. $x \in \mathbb{Z} \rightarrow x \in \mathbb{W}$. This reads the containment $\mathbb{W} \subset \mathbb{Z}$ backward, so expect it to fail. Counterexample: $x = -1$ is an integer but not a whole number. **False.**
2. $x \notin \mathbb{R} \rightarrow x \notin \mathbb{Q}$. This is the contrapositive of $x \in \mathbb{Q} \rightarrow x \in \mathbb{R}$ (#4 above), which is true, so this is **true**. In words: anything off the number line cannot be a fraction, since every fraction is on the number line.
3. $(x \in \mathbb{W} \wedge x \neq 0) \rightarrow x \in \mathbb{N}$. The whole numbers are $0, 1, 2, \dots$, so removing 0 leaves exactly $1, 2, 3, \dots$ **True**. In set language, $\mathbb{W} \setminus \{0\} = \mathbb{N}$.

The pattern to remember: an implication $x \in A \rightarrow x \in B$ that holds for all x is the same thing as $A \subseteq B$, and its contrapositive $x \notin B \rightarrow x \notin A$ says the same thing from the outside in.

6.5 How Big Is an Infinite Set?

We defined the **cardinality** $|A|$ as the number of elements in A , so $|\{1, 2, 3\}| = 3$. What, then, is the cardinality $|\mathbb{N}|$?

Claim: no natural number is the cardinality of $\mathbb{N}$

There is no $N \in \mathbb{N}$ with $|\mathbb{N}| = N$.

The argument works by assuming the opposite and showing that the assumption leads to something impossible (a *proof by contradiction*).

Suppose one were, say $|\mathbb{N}| = N$ for some $N \in \mathbb{N}$. Then $\mathbb{N} = \{1, 2, 3, \dots\}$ would stop after N elements, so its **largest** element would be N itself.

But $N \in \mathbb{N}$ forces $N + 1 \in \mathbb{N}$ as well, and $N + 1 > N$. So N is **not** the largest element, which contradicts what we just said.

Nothing about N was special: *whatever* number we pick, adding 1 produces a bigger one still in $\mathbb{N}$. So the assumption fails for every candidate, and $|\mathbb{N}|$ is no natural number at all. ■

Writing $|\mathbb{N}| = \infty$ does not rescue us either, because we are about to find that **not all infinities are the same size**. We need a notion of “same cardinality” that never requires finishing a count, yet agrees with ordinary counting on finite sets.

6.6 Same Size Means Perfect Pairing

Think of a theater. It has exactly as many seats as ticket-holders when **everyone sits, every seat is filled, and nobody doubles up**. You can confirm that without counting either the seats or the people: the pairing alone settles it.

Definition: Same cardinality (by pairing)

Two sets have the **same cardinality** if their elements can be matched one-to-one, with nothing left over on either side.

For finite sets this agrees with ordinary counting: two finite sets can be paired off perfectly exactly when they have the same number of elements.

Same cardinality

$$A = \{2, 4, 6\}, \quad B = \{a, b, c\}$$

$$\begin{array}{ccc} A & & B \\ 2 & \longleftrightarrow & a \\ 4 & \longleftrightarrow & b \\ 6 & \longleftrightarrow & c \end{array}$$

Nothing left over: $|A| = |B| = 3$.

Different cardinality

$$C = \{1, 2, 3, 4\}, \quad D = \{x, y, z\}$$

$$\begin{array}{ccc} C & & D \\ 1 & \longleftrightarrow & x \\ 2 & \longleftrightarrow & y \\ 3 & \longleftrightarrow & z \\ \boxed{4} & & \text{left over} \end{array}$$

D runs out first, as does every other pairing: $|C| \neq |D|$.

In the second pairing, no amount of rearranging helps: four elements cannot each get their own partner from a set of three. The arrows are double-headed on purpose, because the pairing has to work in both directions: each element on the left gets exactly one partner on the right, and each element on the right gets exactly one partner on the left.

Procedure: checking that a pairing is perfect

1. **Every left element has exactly one partner** on the right.
2. **Every right element is reached**: nothing on the right is left over.
3. **No doubling up**: no right element is the partner of two different left elements.

If all three hold, the two sets have the same cardinality.

Worked example 6.5: Pairings you already use. Each of these is a perfect pairing, and in each case the pairing is how we know the sizes match without counting.

- **Days and numbers.** Pair Monday with 1, Tuesday with 2, and so on to Sunday with 7. Every day gets one number, every number from 1 to 7 is used, and none twice: the days of the week and $\{1, 2, \dots, 7\}$ have the same cardinality. Numbering things *is* pairing them with $\{1, 2, \dots, n\}$; that is what counting means.
- **Patients and beds.** On a full ward with 20 beds and 20 patients, every patient has a bed, every bed has a patient, and nobody shares. That settles $|\text{patients}| = |\text{beds}|$ without counting either.

A **non-example**: 12 patients and 10 beds. However the beds are assigned, once the 10 beds are used, 2 patients are left over. Rearranging who gets which bed never helps, because the sizes really are different.

6.7 The Unsettling Example

The pairing test never asks us to finish counting, so it still works when both sets are **infinite**. Let E be the even natural numbers, and pair each natural number with its double:

$$\begin{array}{ccc} \mathbb{N} = \{1, 2, 3, 4, \dots\} & & E = \{2, 4, 6, 8, \dots\} \\ 1 & \longleftrightarrow & 2 \\ 2 & \longleftrightarrow & 4 \\ 3 & \longleftrightarrow & 6 \\ 4 & \longleftrightarrow & 8 \\ \vdots & & \vdots \\ n & \longleftrightarrow & 2n \end{array}$$

Check the pairing against the definition. The rule $n \leftrightarrow 2n$ reaches **every** even number, since any even number $2n$ is the partner of n . And no natural number is used twice, since different numbers have different doubles. Nothing is left over on either side.

A part can be as big as the whole

So $|\mathbb{N}| = |E|$, even though E is a *proper subset* of $\mathbb{N}$: it is missing every odd number.

For a finite set that is impossible: strip out an element and the count drops. Infinite sets simply do not behave that way.

Worked example 6.6: The odd numbers. Show that the set of odd natural numbers,

$$J = \{1, 3, 5, 7, \dots\},$$

has the same cardinality as $\mathbb{N}$. We need a rule that pairs them perfectly. Listing them in order suggests one: the 1st odd number is 1, the 2nd is 3, the 3rd is 5, so pair

$$n \leftrightarrow 2n - 1 : \quad 1 \leftrightarrow 1, \quad 2 \leftrightarrow 3, \quad 3 \leftrightarrow 5, \quad 4 \leftrightarrow 7, \quad \dots$$

Run the three checks. (1) Every n gets exactly one partner, $2n - 1$, which is odd. (2) Every odd number is reached: which n is paired with 25? Solve $2n - 1 = 25$ to get $n = 13$; in general the odd number m is the partner of $n = \frac{m+1}{2}$, which is a natural number. (3) Different n give different $2n - 1$, so there is no doubling up. So $|J| = |\mathbb{N}|$: there are exactly as many odd numbers as natural numbers.

Worked example 6.7: One extra element changes nothing. The whole numbers $\mathbb{W} = \{0, 1, 2, \dots\}$ are the natural numbers plus one more element, 0. Are they bigger? Pair each natural number with the number one below it:

$$n \leftrightarrow n - 1 : \quad 1 \leftrightarrow 0, \quad 2 \leftrightarrow 1, \quad 3 \leftrightarrow 2, \quad \dots$$

Every whole number w is reached (it is the partner of $n = w + 1$), and no two naturals share a partner. So $|\mathbb{N}| = |\mathbb{W}|$. Everyone “moved over one seat” to make room for 0, and because the list never ends, nobody fell off the end. With a finite list that trick fails: the last person has no seat to move to.

Worked example 6.8: Spreading the numbers out. The multiples of 10, $\{10, 20, 30, \dots\}$, are very sparse: only one natural number in ten belongs. Yet $n \leftrightarrow 10n$ pairs them perfectly with $\mathbb{N}$ (250 is the partner of 25, and in general m is the partner of $\frac{m}{10}$). How spread out the elements are has no effect on cardinality; only whether a perfect pairing exists.

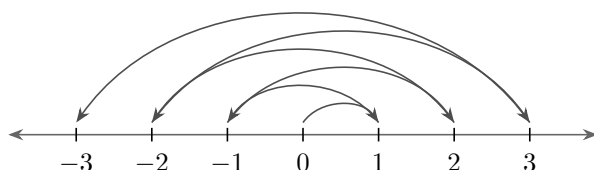
6.8 Countable Infinity: $\aleph_0$

Definition: Countable

A set is **countable** if its elements can be arranged in an infinite **list** $a_1, a_2, a_3, \dots$ reaching every one, that is, if it can be paired off with $\mathbb{N}$.

The pairing is simply position in the list: $1 \leftrightarrow a_1$, $2 \leftrightarrow a_2$, and so on. The size of a countable set is written $\aleph_0$, read “aleph-null” ($\aleph$ is the first letter of the Hebrew alphabet). It is the smallest infinity.

The integers. A list has to start somewhere, and $\mathbb{Z}$ has no smallest element, so we cannot start at $-\infty$ and count upward. Instead, start at 0 and zig-zag outward, alternating between the positive and negative sides:



Reading the arcs in order gives the list

$$0, 1, -1, 2, -2, 3, -3, \dots$$

$$\begin{array}{ccc} \mathbb{N} & & \mathbb{Z} \\ 1 & \longleftrightarrow & 0 \\ 2 & \longleftrightarrow & 1 \\ 3 & \longleftrightarrow & -1 \\ 4 & \longleftrightarrow & 2 \\ \vdots & & \vdots \end{array}$$

Position n in the list pairs with the n th integer visited. Every integer is reached: a positive integer k sits at position $2k$, and a negative integer $-k$ at position $2k + 1$. For example, -2 is 5th, 17 is 34th, and -17 is 35th. No integer is visited twice. So

$$|\mathbb{Z}| = \aleph_0.$$

Worked example 6.9: Finding positions in the list of integers. The list is

$$0, 1, -1, 2, -2, 3, -3, \dots,$$

and the rule from above says: 0 is 1st, a positive integer k is at position $2k$, and a negative integer $-k$ is at position $2k + 1$.

- Where is 5? At position $2 \times 5 = 10$. Where is -5 ? At position $2 \times 5 + 1 = 11$, right after 5.
- Where is 100? At position 200; and -100 at position 201.
- Going the other way, which integer is 30th? An even position $2k$ holds k , so position 30 holds 15. Position 31, odd, is $2 \times 15 + 1$, so it holds -15 .

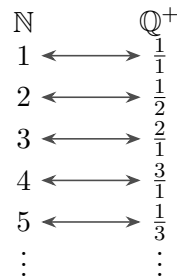
Being able to go both ways (integer to position, position to integer) is exactly what makes the list a perfect pairing: every integer has a position, and every position has an integer.

Watch out: “start at the beginning” does not work for $\mathbb{Z}$

The tempting list $0, 1, 2, 3, \dots$ followed by “then the negatives” never gets to the negatives, because the positives never run out. A list can only reach an element if it appears at some finite position. The zig-zag works because it alternates sides, so every integer, positive or negative, shows up after finitely many steps.

6.9 The Rationals Are Countable Too

Every positive fraction $\frac{p}{q}$ sits in a **grid**: the numerator p picks the row, and the denominator q picks the column. Sweeping the **diagonals** threads one path through all of them.



The fraction $\frac{2}{2}$ is struck out because it *is* $\frac{1}{1}$, which is already on the list. Skipping repeats like this, the path reaches every positive rational exactly once, giving the pairing on the right ($\mathbb{Q}^+$ stands for the positive rationals). Interleaving the negatives and 0 as we did for $\mathbb{Z}$, that is, listing $0, \frac{1}{1}, -\frac{1}{1}, \frac{1}{2}, -\frac{1}{2}, \dots$, gives

$$|\mathbb{Q}| = \aleph_0.$$

- Total 5, the last diagonal in the picture: $\frac{1}{4}, \frac{2}{3}, \frac{3}{2}, \frac{4}{1}$, at positions 6, 7, 8, 9.
- Total 6, running back up: $\frac{5}{1}, \frac{4}{2}, \frac{3}{3}, \frac{2}{4}, \frac{1}{5}$. But $\frac{4}{2} = \frac{2}{1}$, $\frac{3}{3} = \frac{1}{1}$, and $\frac{2}{4} = \frac{1}{2}$ are already on the list, so they are struck out. Only $\frac{5}{1}$ and $\frac{1}{5}$ are new: positions 10 and 11.
- Total 7: $\frac{1}{6}, \frac{2}{5}, \frac{3}{4}, \frac{4}{3}, \frac{5}{2}, \frac{6}{1}$, all in lowest terms, at positions 12 through 17.

So, for example, $\frac{3}{2}$ is 8th on the list and $\frac{5}{2}$ is 16th. Every positive fraction $\frac{p}{q}$ is on the diagonal with total $p + q$, which the path reaches after finitely many steps, so every positive rational gets a position.

6.10 Cantor's Diagonal Argument

The argument again starts by supposing the opposite. Suppose every real between 0 and 1 could be listed too, in *some* order, as $r_1, r_2, r_3, \dots$. Each one has an infinite decimal expansion, so the list

might begin like this:

$$\begin{aligned}
 r_1 &= 0.\underline{2}5819340671\dots \\
 r_2 &= 0.4\underline{9}710283561\dots \\
 r_3 &= 0.60\underline{4}82137952\dots \\
 r_4 &= 0.317\underline{6}05942813\dots \\
 r_5 &= 0.8524\underline{0}3718694\dots \\
 r_6 &= 0.19365\underline{7}240817\dots \\
 r_7 &= 0.472018\underline{3}56921\dots \\
 r_8 &= 0.5083761\underline{9}2470\dots \\
 r_9 &= 0.26194837\underline{5}186\dots \\
 r_{10} &= 0.739162840\underline{2}65\dots \\
 &\vdots
 \end{aligned}$$

The underlined digits run down the **diagonal**: the 1st digit of r_1 , the 2nd digit of r_2 , the 3rd digit of r_3 , and so on.

Build a number the list misses. Walk down the underlined diagonal and replace each digit by **any different** digit: $\underline{2} \rightarrow 7$, $\underline{9} \rightarrow 3$, $\underline{4} \rightarrow 9$, and so on. Writing the new digits one after another gives

$$d = 0.7391628405\dots$$

By construction, the 1st digit of d differs from the 1st digit of r_1 , the 2nd digit of d differs from the 2nd digit of r_2 , and in general the n th digit of d differs from the n th digit of r_n .

The rows may sit in **any order**: nothing here used one. Shuffle them, re-read the new diagonal, and the recipe still works.

Could d be a line further down the list, r_{10} , say? Stack them and the first **nine** digits agree:

$$\begin{aligned}
 d &= 0.739162840\underline{5}\dots \\
 r_{10} &= 0.739162840\underline{2}\dots
 \end{aligned}$$

The tenth digit splits them, since that is the digit the rule changed. Every r_n is defeated the same way at its own n th digit, so d is on **no** line of the list.

One fine point, for the careful reader. A few numbers have two decimal spellings: $0.4999\dots$ (nines forever) and $0.5000\dots$ (zeros forever) are the same number. So a number that differs from r_n in one digit could still equal r_n , if one is the other's second spelling. The careful version of the recipe never uses 0 or 9 as a replacement digit, which rules this out; the example above chose its replacement digits freely, just to show the idea.

Worked example 6.11: The diagonal recipe with a fixed rule. A standard careful rule is: **replace each diagonal digit by 5, unless it already is 5, in which case use 4.** It never

produces a 0 or a 9, and it always changes the digit. Apply it to this (start of a) list:

$$r_1 = 0.\underline{1}41592\dots$$

$$r_2 = 0.5\underline{5}5555\dots$$

$$r_3 = 0.27\underline{1}828\dots$$

$$r_4 = 0.333\underline{3}33\dots$$

$$r_5 = 0.7071\underline{0}6\dots$$

$$r_6 = 0.05000\underline{0}\dots$$

The diagonal digits are 1, 5, 1, 3, 0, 0. The rule turns them into 5, 4, 5, 5, 5, 5, so

$$d = 0.545555\dots$$

Check a few: d differs from r_1 in the 1st digit (5 versus 1), from r_2 in the 2nd digit (4 versus 5), and from r_6 in the 6th digit (5 versus 0). Look closely at $r_2 = 0.555555\dots$ and $d = 0.545555\dots$: they agree everywhere shown except the 2nd digit, and that single digit is enough to make them different numbers. The same holds for every row, however long the list: row n is defeated at digit n .

6.11 What the Diagonal Proves

We assumed the list contained **every** real between 0 and 1. The diagonal hands us one it missed. So no such list exists, and it is not a matter of trying harder, since the recipe defeats *any* list you propose.

If even the reals between 0 and 1 cannot be listed, the whole real line cannot be either. A list of all the reals could be turned into a list of the ones between 0 and 1 just by crossing out the others, and we have shown that no such list exists.

The reals are uncountable

$\mathbb{R}$ cannot be paired off with $\mathbb{N}$. Its cardinality is strictly larger, and is written $\mathfrak{c}$, “the continuum”:

$$\aleph_0 = |\mathbb{N}| = |\mathbb{Z}| = |\mathbb{Q}| < |\mathbb{R}| = \mathfrak{c}$$

Notice where the jump happens. Adding the negatives ($\mathbb{Z}$) and all the fractions ($\mathbb{Q}$) costs nothing. Adding the **irrationals** changes the size of infinity.

6.12 The Gap Between the Two

This leaves an honest open question. Is there a set strictly bigger than $\mathbb{N}$ but strictly smaller than $\mathbb{R}$, some size wedged between $\aleph_0$ and $\mathfrak{c}$?

The guess that there is **no** such size is called the **continuum hypothesis**.

The answer turned out to be stranger than yes or no. Gödel and Cohen proved that the continuum hypothesis can be neither **proved** nor **disproved** from the standard axioms of set theory, the basic starting rules that the rest of set theory is built on. Both answers are consistent with those rules, meaning neither one leads to a contradiction. You may assume either one.

It is a reminder that “how many?” is a deeper question than it looks, and that the number line you have used since grade school still holds open problems.

6.13 Extra Practice

Exercise 6.3

Classify each number: which of $\mathbb{N}, \mathbb{W}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ contain it?

$$-\sqrt{25}, \quad 1.25, \quad 0.666\dots, \quad \sqrt{\frac{1}{4}}, \quad 3\pi, \quad \frac{100}{4}$$

Solution to Exercise 6.3

Simplify each one first, then find its smallest set.

- $-\sqrt{25} = -5$: a negative integer. In $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$.
- $1.25 = \frac{125}{100} = \frac{5}{4}$: a ratio of integers, not whole. In $\mathbb{Q}, \mathbb{R}$.
- $0.666\dots = \frac{2}{3}$ (the decimal repeats, so it is rational; long division of 2 by 3 gives 6’s forever). In $\mathbb{Q}, \mathbb{R}$.
- $\sqrt{\frac{1}{4}} = \frac{1}{2}$, since $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$. Another square root that is not irrational. In $\mathbb{Q}, \mathbb{R}$.
- 3π : irrational. (If 3π were a fraction $\frac{p}{q}$, then π would be $\frac{p}{3q}$, also a fraction, which it is not.) In $\mathbb{R}$ only.
- $\frac{100}{4} = 25$: a natural number. In all five.

Exercise 6.4

Give a perfect pairing between $\mathbb{N}$ and the multiples of 5, $\{5, 10, 15, \dots\}$. Which natural number is paired with 85? What does the pairing show?

Solution to Exercise 6.4

Pair $n \leftrightarrow 5n$: $1 \leftrightarrow 5$, $2 \leftrightarrow 10$, $3 \leftrightarrow 15$, and so on. Every n gets one partner; every multiple of 5, say m , is reached as the partner of $\frac{m}{5}$; and different n have different partners. The number paired with 85 is $\frac{85}{5} = 17$. The pairing shows that the multiples of 5 have cardinality $\aleph_0$, the same as $\mathbb{N}$, even though they form a proper subset of $\mathbb{N}$.

Exercise 6.5

In the list $0, 1, -1, 2, -2, 3, -3, \dots$ of the integers, at which position does -8 appear? Which integers are at positions 12 and 13?

Solution to Exercise 6.5

A negative integer $-k$ is at position $2k + 1$, so -8 is at position $2 \times 8 + 1 = 17$. An even position $2k$ holds k , so position $12 = 2 \times 6$ holds 6; the odd position $13 = 2 \times 6 + 1$ holds -6 . You can confirm by writing the list out: $0, 1, -1, 2, -2, 3, -3, 4, -4, 5, -5, 6, -6$ has 6 in the 12th slot and -6 in the 13th.

Exercise 6.6

Use the rule “replace each diagonal digit by 5, unless it is 5, in which case use 4” to build a number d between 0 and 1 that is missing from this list:

$$r_1 = 0.31415\dots$$

$$r_2 = 0.25182\dots$$

$$r_3 = 0.14142\dots$$

$$r_4 = 0.50000\dots$$

$$r_5 = 0.66666\dots$$

Explain why d is not r_4 .

Solution to Exercise 6.6

The diagonal digits are the 1st digit of r_1 (3), the 2nd of r_2 (5), the 3rd of r_3 (1), the 4th of r_4 (0), and the 5th of r_5 (6). The rule changes them to 5, 4, 5, 5, 5, so

$$d = 0.54555\dots$$

Compare with $r_4 = 0.50000\dots$: the 4th digit of d is 5, and the 4th digit of r_4 is 0. (They differ at the 2nd, 3rd, and 5th digits too, but the 4th is the one the recipe guarantees, whatever the rest of r_4 looks like.) Because the rule never uses 0 or 9, d has only one decimal spelling, so differing in a digit really does make it a different number. The same reasoning rules out every r_n at its own n th digit.

7 Intervals

7.1 Intervals, and How We Write Them

In Exercise 2.1, “the numbers between 10 and 20” left it unclear whether 10 and 20 themselves count. Intervals are the notation that settles that question.

Definition: Interval

An **interval** is the set of **all** real numbers lying between two given numbers a and b , called the **endpoints**. The bracket records whether each endpoint is itself included:

$$[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$$

$$(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$$

$$[a, b) = \{x \in \mathbb{R} \mid a \leq x < b\}$$

$$(a, b] = \{x \in \mathbb{R} \mid a < x \leq b\}$$

A square bracket means the endpoint is included, and a round bracket means it is not. The four kinds have names:

- **Closed:** $[a, b]$, **both** endpoints included.
- **Open:** (a, b) , **neither** endpoint included.
- **Half-open:** $[a, b)$ or $(a, b]$, one of each (also called **half-closed**).

Note that an interval contains *every* real number in its stretch, not just the whole numbers: $[2, 7]$

contains 2.5, π , and 6.999, not only 2, 3, 4, 5, 6, 7.

Intuitively, an interval is an **unbroken stretch** of the number line: if two numbers are in it, so is every number between them. That is exactly the shape of most clinical cutoffs.

- A study enrolling adults “aged 18 to 65 inclusive” accepts ages in $[18, 65]$.
- A temperature of 38.0°C or higher is commonly treated as a fever: the fever range is $[38, \infty)$, an interval with no upper end (we meet these in a moment).
- A normal fasting blood glucose is often quoted as 70 to 99 mg/dL. Since a measured value can fall between whole numbers, the precise version is “at least 70 and below 100,” the interval $[70, 100)$.

Procedure: is a number in an interval?

Translate the interval into its two inequalities (one per endpoint) and test the number against both. Square bracket means $\leq$ (or $\geq$); round bracket means $<$ (or $>$). The number is in the interval only if it passes both tests.

Worked example 7.1: Membership in intervals. Decide each.

1. Is $2 \in [2, 5)$? The tests are $2 \leq x$ and $x < 5$. $2 \leq 2$ is true (the square bracket allows equality) and $2 < 5$ is true. **Yes.**
2. Is $5 \in [2, 5)$? $2 \leq 5$ is true, but $5 < 5$ is false. **No:** the round bracket keeps the endpoint out.
3. Is $4.999 \in [2, 5)$? $2 \leq 4.999$ and $4.999 < 5$: both true. **Yes.** Numbers can get as close to 5 as you like and still be in; only 5 itself is excluded.
4. Is $0 \in (0, 1)$? $0 < 0$ is false. **No.**
5. Is a fasting glucose of 99.6 mg/dL in the normal range $[70, 100)$? $70 \leq 99.6$ and $99.6 < 100$: **yes.** Is 100? No: $100 < 100$ fails, so a reading of exactly 100 is outside the normal range.

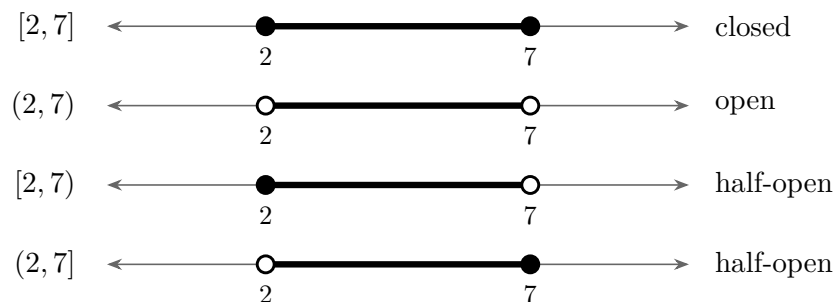
Non-examples. The set $\{2, 3, 4, 5, 6, 7\}$ is *not* the interval $[2, 7]$: it has gaps (it misses 2.5, for instance). Likewise, “the number of doses a patient took this week,” which can only be $0, 1, 2, \dots, 7$, is a set of whole numbers, not an interval. Intervals are for quantities that can take *any* value in a stretch, like weight, time, or a lab concentration.

Watch out: write the smaller endpoint first

An interval is always written $[a, b]$ with $a \leq b$: left end first, as on the number line. “The numbers between 2 and 5” is $(2, 5)$, never $(5, 2)$. Two small consequences of the definitions: $[a, a] = \{a\}$ (only a satisfies $a \leq x \leq a$), while $(a, a) = \emptyset$ (no number satisfies $a < x < a$).

7.2 One Decision, Two Endpoints, Four Intervals

Each endpoint decides on its own whether it is in or out, so two endpoints give $2 \times 2 = 4$ intervals. Here they are on the endpoints 2 and 7:



How to read these pictures: the thin line with an arrowhead at each end is the number line, which runs forever in both directions. The thick band marks the numbers in the interval. A **filled** dot means that endpoint is included, and an **open** (hollow) dot means it is not.

Only the endpoints differ: every number strictly between 2 and 7 belongs to all four. And the three notations (bracket, dot, and inequality) agree endpoint by endpoint:

Bracket	Dot	Inequality	The endpoint itself
square, []	filled ●	$\leq, \geq$ (bar)	is in the set
round, ()	open ○	$<, >$ (no bar)	is not in the set

The “bar” is the extra line under $\leq$ and $\geq$, which allows equality. So a square bracket, a filled dot, and an inequality with a bar all say the same thing: the endpoint is included.

Everyday English has its own words for the two choices, and it pays to learn them as pairs:

Endpoint included	Endpoint excluded	In symbols
“at least a ,” “ a or more”	“more than a ,” “above a ”	$x \geq a$ versus $x > a$
“at most b ,” “no more than b ”	“less than b ,” “under b ”	$x \leq b$ versus $x < b$
“from a to b inclusive”	“strictly between a and b ”	$[a, b]$ versus (a, b)

“Between” on its own is the ambiguous one, as Exercise 2.1 found. When you write a rule for others, say “inclusive” or “strictly” explicitly.

7.3 Every Interval, Three Ways

An interval does not have to stop on both sides. Using ∞ (“infinity”) for a side with no end, there are nine kinds in all. The first four are **bounded**: they have two endpoints and stop in both directions. The rest are **unbounded**: they run off forever in at least one direction, which the picture shows as the thick band running out to the arrowhead.

Interval	Number line	Set-builder
(a, b)		$\{x \in \mathbb{R} \mid a < x < b\}$
$(a, b]$		$\{x \in \mathbb{R} \mid a < x \leq b\}$
$[a, b)$		$\{x \in \mathbb{R} \mid a \leq x < b\}$
$[a, b]$		$\{x \in \mathbb{R} \mid a \leq x \leq b\}$
(a, ∞)		$\{x \in \mathbb{R} \mid x > a\}$
$[a, \infty)$		$\{x \in \mathbb{R} \mid x \geq a\}$
$(-\infty, b)$		$\{x \in \mathbb{R} \mid x < b\}$
$(-\infty, b]$		$\{x \in \mathbb{R} \mid x \leq b\}$
$(-\infty, \infty)$		$\mathbb{R}$

∞ is **not a number**, just shorthand for “without end,” so its side always takes a **round** bracket. There is no endpoint out there to include. Notice also that the set-builder form of an unbounded interval has an inequality only on a side that has an endpoint.

Procedure: translating a description into an interval

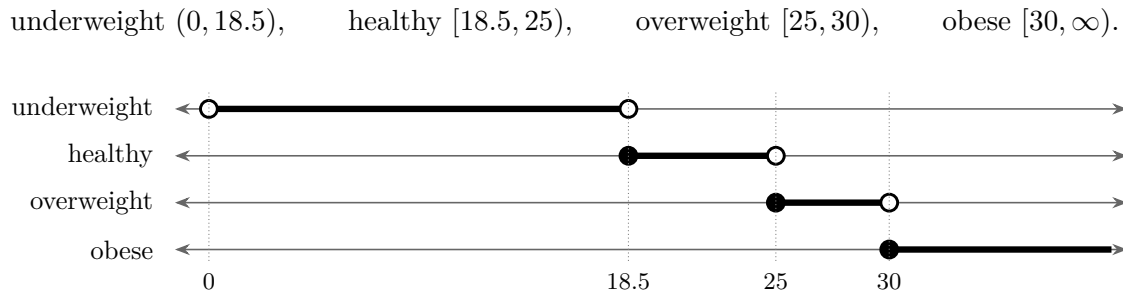
1. Find the lower limit and the upper limit, if any. A side with no limit gets $-\infty$ or ∞ .
2. For each limit separately, decide whether the limit value itself is allowed (square bracket, $\leq$) or not (round bracket, $<$).
3. Write the interval left end first, and check with one number just inside and one just outside.

Worked example 7.2: From words to intervals. Translate each description into interval notation and an inequality.

1. **“Aged at least 18 but under 65.”** Lower limit 18, allowed (“at least”); upper limit 65, not allowed (“under”). So $[18, 65)$, that is, $18 \leq x < 65$. Check: someone aged 64.9 is in, someone aged exactly 65 is out.
2. **“Symptoms lasting more than 2 hours.”** Lower limit 2, not allowed (“more than”); no upper limit. So $(2, \infty)$, that is, $x > 2$.
3. **“A systolic pressure no higher than 140.”** No lower limit written; upper limit 140, allowed. So $(-\infty, 140]$, that is, $x \leq 140$. (Real pressures are positive, of course; the interval just records the one limit the description gives.)
4. **“Within 3 units of 10.”** “Within” means the distance from 10 is at most 3, so x can go down to $10 - 3 = 7$ and up to $10 + 3 = 13$, both allowed. So $[7, 13]$, that is, $7 \leq x \leq 13$.
5. **“ $x \leq 7$ and $x > -2$.”** The two inequalities are given in an awkward order. Rewrite the second as $-2 < x$ and put the lower limit first: $-2 < x \leq 7$, which is $(-2, 7]$.

6. “ $-4 < x$.” Read it from x ’s point of view: $x > -4$. So $(-4, \infty)$. A common slip is to read “ $-4 <$ ” as “less than -4 ” and write $(-\infty, -4)$, which is exactly backward.

Worked example 7.3: Half-open bins for clinical categories. Adult BMI (body mass index) is sorted into four categories: underweight below 18.5, healthy weight from 18.5 to less than 25, overweight from 25 to less than 30, and obesity at 30 or higher. As intervals (BMI is always positive):



Why half-open? Look at the boundary value 25. It must belong to exactly one category. The healthy bin $[18.5, 25)$ leaves it out, and the overweight bin $[25, 30)$ takes it in, so a BMI of exactly 25.0 is “overweight,” with no argument possible. Now imagine using closed intervals $[18.5, 25]$ and $[25, 30]$: then 25 is in *both*, and a patient could be counted twice. With open intervals $(18.5, 25)$ and $(25, 30)$, 25 is in *neither*, and a patient falls through the crack. Half-open bins, all closed on the same side, fit together edge to edge with no overlaps and no gaps, so every value lands in exactly one bin.

Classifying a few values: 18.4 is underweight; 18.5 is healthy (the square bracket takes it); 24.9 is healthy; 25.0 is overweight; 29.99 is overweight; 30.0 and 32.7 are obese.

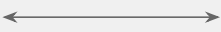
The same design shows up in software. When NumPy’s `histogram` function sorts values into bins with edges 1, 2, 3, 4, the bins are $[1, 2)$, $[2, 3)$, and $[3, 4)$: each bin includes its left edge, and only the last bin also includes its right edge, so that the largest value is not lost. A value of exactly 2 goes into the second bin, not the first.

Watch out: “a to b” categories on whole numbers

Guidelines are often written with whole numbers: “elevated blood pressure is a systolic reading of 120 to 129.” Read as a closed interval $[120, 129]$, that leaves a gap: where does 129.5 go? The guideline’s intent is the half-open interval $[120, 130)$, since it was written for whole-number readings. When you turn such categories into intervals for continuous data, check that consecutive bins meet edge to edge.

Exercise 7.1

Each row is filled in *one* way. Fill in the other three. Watch each endpoint separately: they are independent decisions.

	Number line	Interval	Inequality	Set-builder
1.		$[-3, 5)$		
2.			$-1 < x \leq 6$	
3.				$\{x \in \mathbb{R} \mid x \geq 4\}$
4.				

Solution to Exercise 7.1

In each row, read off the two endpoints and decide, one at a time, whether each is included. That single decision fixes the bracket, the dot, and the inequality symbol together.

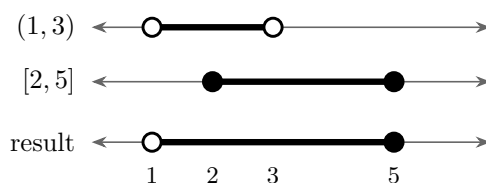
	Number line	Interval	Inequality	Set-builder
1.		$[-3, 5)$	$-3 \leq x < 5$	$\{x \in \mathbb{R} \mid -3 \leq x < 5\}$
2.		$(-1, 6]$	$-1 < x \leq 6$	$\{x \in \mathbb{R} \mid -1 < x \leq 6\}$
3.		$[4, \infty)$	$x \geq 4$	$\{x \in \mathbb{R} \mid x \geq 4\}$
4.		$(-\infty, 0)$	$x < 0$	$\{x \in \mathbb{R} \mid x < 0\}$

In #1 the square bracket at -3 means -3 is included (filled dot, $\leq$), while the round bracket at 5 means 5 is not (open dot, $<$). #2 is the mirror image: open at -1 , closed at 6 . #3 and #4 are unbounded, so the infinite side takes a **round** bracket: there is no endpoint there to include. In #4 the open dot at 0 is what makes it $<$ rather than $\leq$.

7.4 Union of Intervals

Intervals are sets, so the operations of Section 4 apply to them. To combine two intervals, draw both, then read off the answer. For a union, the answer is everything covered by **either** one.

Bounded example: $(1, 3) \cup [2, 5] = (1, 5]$.



The two intervals overlap (on the stretch from 2 to 3), so they merge into one stretch: open at 1, closed at 5. The left end is open because 1 is in neither interval; the right end is closed because 5 is in $[2, 5]$, and being in one of them is enough for a union.

Unbounded example: $(-\infty, 2] \cup (3, \infty)$, a union that does *not* merge.



There is no overlap, so nothing merges: the numbers after 2, up to and including 3, are in neither piece. The union is genuinely two pieces, and this is already its simplest form. An “abnormal lab value” set looks exactly like this: too low, or too high.

Procedure: the union of two intervals

1. Draw both intervals on stacked number lines.
2. If they overlap, or meet at a point that belongs to at least one of them, they **merge** into a single interval running from the leftmost left end to the rightmost right end.
3. Each end of the result keeps the bracket of the interval that supplied it. If both intervals share that end, it is included if *either* one includes it.
4. If there is a gap between them, even a single missing point, the union stays as two pieces joined by $\cup$.

Worked example 7.4: Touching intervals: merge or not? Compare $[0, 2) \cup [2, 5]$ with $(0, 2) \cup (2, 5)$.



On the left, the two intervals meet at 2, and 2 belongs to $[2, 5]$. So there is no gap: every number from 0 to 5 is covered, and $[0, 2) \cup [2, 5] = [0, 5]$. The left end 0 comes from $[0, 2)$ with a square bracket, and the right end 5 from $[2, 5]$ with a square bracket.

On the right, the two intervals also meet at 2, but 2 belongs to *neither*. The union covers everything from 0 to 5 except the single point 2, a one-point hole (the open dot in the middle of the result). So $(0, 2) \cup (2, 5)$ is *not* $(0, 5)$; it cannot be written as a single interval at all, and $(0, 2) \cup (2, 5)$ is already its simplest form. A single missing point is enough to break an interval in two.

Worked example 7.5: One interval inside another. Find $[-1, 4) \cup (0, 2]$. Draw them: $(0, 2]$ lies entirely inside $[-1, 4)$, since $-1 \leq 0$ and $2 < 4$. Adding a piece that was already covered changes nothing, so the union is $[-1, 4)$. In general, if $A \subseteq B$ then $A \cup B = B$.

Worked example 7.6: Unbounded meets bounded. Find $(-\infty, -1] \cup [-3, 2)$. The first interval covers everything up to -1 ; the second covers -3 up to (not including) 2. They overlap on $[-3, -1]$, so they merge. The leftmost end is $-\infty$, from the first interval; the rightmost end is 2, from the second, with its round bracket. So the union is $(-\infty, 2)$.

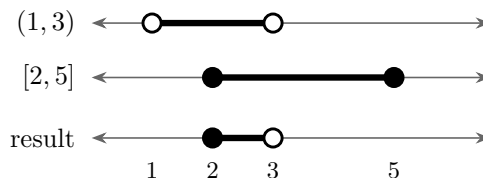
Worked example 7.7: Combining BMI categories. “Overweight or healthy weight” is

$[18.5, 25) \cup [25, 30)$. The two bins meet at 25, which belongs to the second one, so they merge: the union is $[18.5, 30)$, “a BMI of at least 18.5 and below 30.” Because the bins were designed to fit edge to edge, neighboring categories always merge cleanly like this.

7.5 Intersection of Intervals

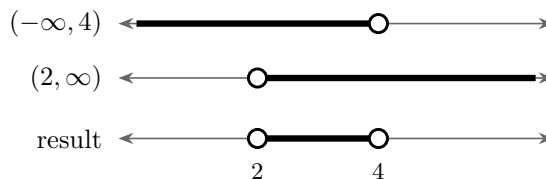
For an intersection, only the stretch covered by **both** intervals survives.

Bounded example: $(1, 3) \cap [2, 5] = [2, 3)$.



Only the overlap survives. 2 lies in **both** intervals, so it is included. 3 lies in $[2, 5]$ but not in $(1, 3)$, so it is not in both, and it is left out.

Unbounded example: $(-\infty, 4) \cap (2, \infty) = (2, 4)$.



Two infinite sets can meet in a bounded one. Neither endpoint is in both, so both stay open: 2 is in $(-\infty, 4)$ but not in $(2, \infty)$, and 4 is in $(2, \infty)$ but not in $(-\infty, 4)$.

Procedure: the intersection of two intervals

1. The left end of the result is the **larger** of the two left ends; the right end is the **smaller** of the two right ends. (Both conditions must hold, so the stricter limit wins on each side.)
2. An end is included only if it belongs to **both** intervals.
3. If the left end comes out bigger than the right end, the intervals do not overlap and the intersection is $\emptyset$.

Worked example 7.8: Two bounded intervals. Find $[0, 5] \cap [3, 8]$. The left ends are 0 and 3; the larger is 3. The right ends are 5 and 8; the smaller is 5. Is 3 in both? 3 is in $[0, 5]$ and in $[3, 8]$: yes, so it is included. Is 5 in both? Yes. So $[0, 5] \cap [3, 8] = [3, 5]$.

Worked example 7.9: Mixed brackets. Find $[-2, 6) \cap (1, 9]$. Larger left end: 1, which comes from $(1, 9]$ and is not in it, so it is excluded. Smaller right end: 6, which comes from $[-2, 6)$ and is not in it, so it is excluded. The result is $(1, 6)$. Notice that -2 and 9 played no part: an intersection’s ends are always the “inner” ends.

Worked example 7.10: Touching at a single point. Compare $[1, 3) \cap [3, 6]$ with $[1, 3] \cap [3, 6]$.



The two pairs of intervals look almost identical, and the answers are completely different in kind. On the left, the only candidate for a shared point is 3, and 3 is not in $[1, 3)$. So nothing is in both: $[1, 3) \cap [3, 6] = \emptyset$. On the right, 3 is in both, so $[1, 3] \cap [3, 6] = \{3\}$, a single point (which is also the interval $[3, 3]$). A whole answer can hinge on one bracket.

Worked example 7.11: Combining eligibility criteria. A trial enrolls adults aged 18 to 65 inclusive, $[18, 65]$. A sub-study within it requires age at least 40, $[40, \infty)$. Which ages qualify for the sub-study? Both conditions must hold, so intersect:

$$[18, 65] \cap [40, \infty) = [40, 65].$$

The larger left end is 40 (in both, so included); the smaller right end is 65 (in both, so included). Every eligibility list joined by “and” is an intersection, and with numeric criteria it is usually an intersection of intervals.

Worked example 7.12: No overlap at all. Find $(-\infty, 1) \cap (3, \infty)$. Larger left end: 3. Smaller right end: 1. The left end is bigger than the right end, so there is nothing in between: the intersection is $\emptyset$. No number is both less than 1 and greater than 3.

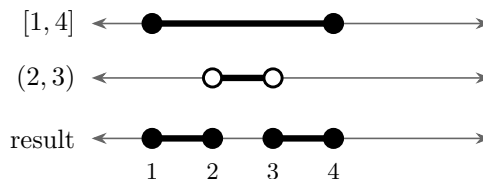
Watch out: union and intersection handle shared endpoints oppositely

For a **union**, a shared endpoint is included if *either* interval includes it (being in one is enough). For an **intersection**, it is included only if *both* do. This is just OR versus AND again, applied to a single point, and it is where most interval mistakes happen.

7.6 Difference and Complement

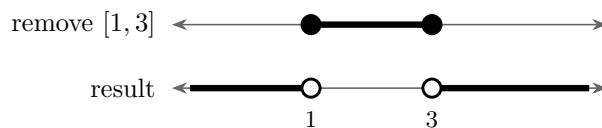
The difference $A \setminus B$ keeps what is in A but **not** in B . With the real line as the universe, the **complement** of an interval A is the case $\mathbb{R} \setminus A$: everything on the line outside A .

Bounded example: $[1, 4] \setminus (2, 3) = [1, 2] \cup [3, 4]$.



Removing an *open* middle leaves two *closed* pieces. The numbers 2 and 3 were never in $(2, 3)$, so they were never removed, and they stay behind as the inner endpoints of the two pieces.

Unbounded example: $\mathbb{R} \setminus [1, 3] = (-\infty, 1) \cup (3, \infty)$.

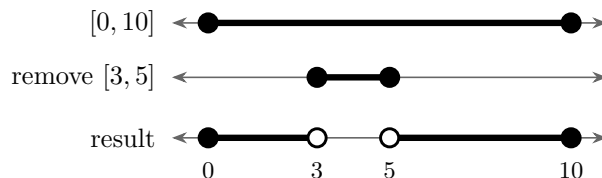


Removing a *closed* interval leaves *open* ends: 1 and 3 were inside $[1, 3]$, so they are gone. The two cases are mirror images, since whatever an endpoint's status in the removed piece, it gets the opposite status in what remains. **Watch the endpoints:** every mistake happens there.

Procedure: removing an interval B from an interval A

1. Draw A , and mark the part of it covered by B .
2. Keep what is left of A : nothing, one piece, or two pieces (if B cuts out a middle section).
3. At each place where B cut A , **flip** B 's bracket: a point B included is removed, so the remainder is open there; a point B excluded stays behind, so the remainder is closed there.
4. The ends of A that B did not touch keep A 's own brackets.

Worked example 7.13: Cutting a closed piece out of the middle. Find $[0, 10] \setminus [3, 5]$.



$[3, 5]$ lies inside $[0, 10]$, so removing it leaves two pieces. The cuts are at 3 and 5, both included in $[3, 5]$, so both are removed: the remainder is open there. The outer ends 0 and 10 were untouched and keep their square brackets. So

$$[0, 10] \setminus [3, 5] = [0, 3) \cup (5, 10].$$

Worked example 7.14: Changing one bracket. Now remove $(3, 5]$ instead: $[0, 10] \setminus (3, 5]$. The cut at 3 used a round bracket, so 3 was not removed and stays behind; the cut at 5 used a square bracket, so 5 goes. The result is $[0, 3] \cup (5, 10]$. Compared with the previous example, one bracket in the removed piece flipped one bracket in the answer.

Worked example 7.15: Removing a piece that hangs off the end. Find $[2, 8) \setminus [5, 12]$. The removed interval covers the right part of $[2, 8)$, from 5 onward (and continues past 8, which does not matter: you cannot remove what was never there). What remains is a single piece from 2 up to the cut at 5. 5 is in $[5, 12]$, so it is removed: $[2, 8) \setminus [5, 12] = [2, 5)$.

Worked example 7.16: Abnormal lab values. With a normal fasting glucose range of $[70, 100)$ mg/dL, the abnormal readings are its complement on the real line:

$$\mathbb{R} \setminus [70, 100) = (-\infty, 70) \cup [100, \infty).$$

Flip each bracket: 70 was in the normal range, so the “too low” piece stops short of it, $(-\infty, 70)$; 100 was not in the normal range, so the “too high” piece includes it, $[100, \infty)$. A reading of exactly 100 is flagged; a reading of exactly 70 is not. (Real glucose readings are positive, so in practice the low piece is $(0, 70)$; the complement on $\mathbb{R}$ is the pure-math version.)

Worked example 7.17: De Morgan’s law on intervals. Find the complement of the two-piece set $(-\infty, 2] \cup (3, \infty)$ from the union subsection. By De Morgan, the complement of a union is the intersection of the complements:

$$\overline{(-\infty, 2] \cup (3, \infty)} = \overline{(-\infty, 2]} \cap \overline{(3, \infty)} = (2, \infty) \cap (-\infty, 3] = (2, 3].$$

Each complement flips a bracket (2 was in, so its complement starts just after it; 3 was out, so its complement includes it), and the intersection keeps the inner ends. This matches the picture: the union had a gap from just after 2 up to and including 3, and the complement is exactly that gap.

Watch out: a difference can leave a single point

$[1, 5] \setminus [1, 5)$ removes everything except 5, since 5 is in $[1, 5]$ but not in $[1, 5)$. The answer is $\{5\}$, not $\emptyset$. Whenever two intervals differ only in a bracket, their difference is the endpoint where they differ.

7.7 Extra Practice

Exercise 7.2

Let $A = [-4, 1)$ and $B = [0, 3]$. Find $A \cup B$, $A \cap B$, $A \setminus B$, and $B \setminus A$.

Solution to Exercise 7.2

Draw them first: A runs from -4 (included) to 1 (excluded), B from 0 to 3 (both included). They overlap on the stretch from 0 to 1 .

- $A \cup B$: they overlap, so they merge, from the leftmost end -4 (square, from A) to the rightmost end 3 (square, from B): $[-4, 3]$.
- $A \cap B$: larger left end 0 (in both, so included); smaller right end 1 (not in A , so excluded): $[0, 1)$.
- $A \setminus B$: remove $[0, 3]$ from A . The cut is at 0 , which B includes, so 0 is removed: $[-4, 0)$.
- $B \setminus A$: remove $[-4, 1)$ from B . The cut is at 1 , which A excludes, so 1 stays behind: $[1, 3]$.

Check: $A \setminus B$, $A \cap B$, and $B \setminus A$ are $[-4, 0)$, $[0, 1)$, $[1, 3]$, which fit edge to edge with no overlaps and together make $[-4, 3] = A \cup B$, exactly like the three regions inside a two-set Venn diagram.

Exercise 7.3

Write the complement of $(-2, 5]$ in $\mathbb{R}$, and then find $(-\infty, 3) \cap [3, \infty)$ and $(-\infty, 3) \cup [3, \infty)$.

Solution to Exercise 7.3

Complement. Everything outside $(-2, 5]$, flipping both brackets: -2 was excluded, so it now

belongs to the left piece; 5 was included, so it is now excluded from the right piece:

$$\mathbb{R} \setminus (-2, 5] = (-\infty, -2] \cup (5, \infty).$$

Intersection. The only candidate for a shared point is 3, and $3 \notin (-\infty, 3)$. So $(-\infty, 3) \cap [3, \infty) = \emptyset$.

Union. The two meet at 3, which belongs to the second, so there is no gap: $(-\infty, 3) \cup [3, \infty) = (-\infty, \infty) = \mathbb{R}$. Together these two facts say that the pair splits the real line cleanly into two bins, with every number in exactly one: the same edge-to-edge design as the BMI categories.

Exercise 7.4

Looking at the systolic number alone, a blood pressure reading is often classed as normal below 120, elevated from 120 to 129, stage 1 hypertension from 130 to 139, and stage 2 hypertension at 140 or above. (The full guideline also uses the diastolic number.) Write the four classes as half-open intervals that fit edge to edge, and classify readings of 118, 120, 129.5, 130, and 140.

Solution to Exercise 7.4

The guideline’s “120 to 129” was written for whole-number readings; for continuous values it means “at least 120 and below 130.” Treating pressures as positive:

normal $(0, 120)$, elevated $[120, 130)$, stage 1 $[130, 140)$, stage 2 $[140, \infty)$.

Each boundary value belongs to the class on its right, and there are no gaps. Classifying: 118 is normal; 120 is elevated (the square bracket of $[120, 130)$ takes it); 129.5 is elevated (a reading the whole-number version could not place); 130 is stage 1; 140 is stage 2.

8 Ordered Pairs and Tuples

8.1 When Order Matters

In a set, order is irrelevant: $\{2, 5\} = \{5, 2\}$. Sometimes that is exactly wrong.

Definition: Ordered pair

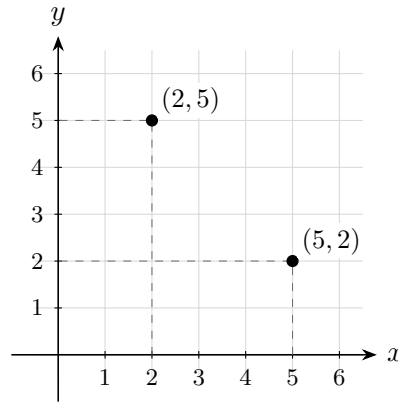
An **ordered pair** (a, b) is a collection of two elements in which **the order matters**:

$$(a, b) = (c, d) \iff a = c \text{ and } b = d$$

Read the definition as: two ordered pairs are equal if and only if their first entries match *and* their second entries match. Having the same two numbers in some order is not enough; each number has to be in the same slot.

The one you already know: points on an xy graph. Every point you have ever plotted is an ordered pair, with the x -coordinate first and the y -coordinate second. So $(2, 5)$ means *right 2, up 5*, while $(5, 2)$ means *right 5, up 2*: two different places from the same two numbers.

A notation warning. $(2, 5)$ is also how we write an open interval. **Context tells you which is meant:** a pair of coordinates, or a stretch of the number line.



The same two numbers, in the two possible orders, give two different points. Coordinates are the clearest case of a general idea: an ordered pair has a **first slot** and a **second slot**, and each slot has its own meaning.

Examples from health data.

- **Blood pressure.** A reading of “120 over 80” is the pair (systolic, diastolic) = (120, 80). The pair (80, 120) would claim a systolic pressure of 80 and a diastolic of 120, a completely different (and physiologically alarming) reading. As a set, $\{120, 80\} = \{80, 120\}$, and the information about which number is which is lost.
- **Dates.** In the United States, (3, 4) as (month, day) is March 4, while (4, 3) is April 3. Many countries write the day first, which is why dates exchanged between hospital systems are a classic source of data errors: the *values* are right but the *slots* were read in the wrong order.
- **Height and weight.** The pair (170, 70) recorded as (height in cm, weight in kg) describes a very different person from (70, 170).

In each case the non-example, the set, fails for the same reason: a set cannot tell its elements apart by position, because it has no positions.

Worked example 8.1: Solving an equation between ordered pairs. Find x and y if $(x + 1, 2y) = (5, 8)$. Two ordered pairs are equal exactly when the first entries match and the second entries match, so one pair equation is really two ordinary equations:

$$x + 1 = 5 \quad \text{and} \quad 2y = 8.$$

So $x = 4$ and $y = 4$. Check: $(4 + 1, 2 \times 4) = (5, 8)$.

Worked example 8.2: Entries that depend on each other. Find a and b if $(2a, b - 3) = (b, 1)$. Match slot by slot: $2a = b$ and $b - 3 = 1$. The second equation involves only b , so solve it first: $b = 4$. Then the first gives $2a = 4$, so $a = 2$. Check: $(2 \times 2, 4 - 3) = (4, 1)$, and the right side is $(b, 1) = (4, 1)$. They agree.

Worked example 8.3: An equation with no solution. Is there an a with $(a, 3) = (4, a)$? Slot by slot: $a = 4$ from the first entries, and $3 = a$ from the second. Both cannot hold, since a cannot be 4 and 3 at once. So **no** value of a works. Slot-by-slot matching can give too many conditions to satisfy at once; that is a legitimate answer, not a mistake.

Worked example 8.4: The same equation with sets instead of pairs. Now find all x, y with $\{x + 1, 2y\} = \{5, 8\}$, sets rather than ordered pairs. A set does not care about order, so the two elements can match up either way:

- $x + 1 = 5$ and $2y = 8$, giving $x = 4, y = 4$; or
- $x + 1 = 8$ and $2y = 5$, giving $x = 7, y = 2.5$.

(If $x + 1$ and $2y$ were equal to each other, the left side would be a set with one element, which cannot equal the two-element set $\{5, 8\}$.) So the set equation has **two** solutions, while the ordered-pair equation had exactly one. Order is information: the pair tells you which number goes where, and the set does not.

Watch out: tuples keep repeats, sets do not

$(1, 1)$ is a perfectly good ordered pair: the point on the diagonal line one step right and one step up. But $\{1, 1\}$ is just the set $\{1\}$. A patient whose systolic and diastolic readings happened to be equal would still have a two-slot reading; the slots exist whether or not their contents differ.

8.2 Tuples

Definition: n -tuple

An **n -tuple** $(a_1, a_2, \dots, a_n)$ generalizes the ordered pair to any finite number of entries, order still mattering.

The small numbers set low beside the letters are **subscripts**: a_1 is the first entry, a_2 the second, and a_n the last. Some tuples have their own names:

- A **2-tuple** is an ordered pair: (a, b) .
- A **3-tuple** is a triple: (a, b, c) .
- A **4-tuple**: (a, b, c, d) . And so on.

Two n -tuples are equal exactly when they agree entry by entry: $a_i = b_i$ for every i . Here i stands for any position from 1 to n , so this says the first entries match, the second entries match, and so on through the last.

For example:

- $(2, 5) \neq (5, 2)$: order matters.
- $(1, 2, 3) \neq (3, 2, 1)$, for the same reason.
- $(x, y, z) = (4, -2, 7)$ locates a point in three-dimensional space: 4 along the first axis, -2 along the second, and 7 along the third.

A patient record is a tuple. Each row of the study table in Section 1.3 is a 7-tuple, one entry per column, in the column order (patient, age, sex, diabetic, hypertensive, obese, pregnant). Patient 4's row is

$(P4, 66, M, \text{yes}, \text{yes}, \text{yes}, \text{no})$.

The position of each entry is what tells you what it means: the 66 is an age *because* it sits in the second slot. Swap two entries and the tuple describes a different (and possibly nonsensical) patient.

Worked example 8.5: Equality of triples. Find x, y, z with $(x, y + 2, 3z) = (-1, 5, 12)$. Match entry by entry: $x = -1$; $y + 2 = 5$, so $y = 3$; $3z = 12$, so $z = 4$. Three slots give three equations. Check: $(-1, 3 + 2, 3 \times 4) = (-1, 5, 12)$.

Sets and tuples side by side.

	Set	Tuple
Brackets	curly: $\{1, 1, 2\}$	round: $(1, 1, 2)$
Does order matter?	no: $\{1, 2\} = \{2, 1\}$	yes: $(1, 2) \neq (2, 1)$
Do repeats matter?	no: $\{1, 1, 2\} = \{1, 2\}$	yes: $(1, 1, 2) \neq (1, 2)$
Size of the example	$ \{1, 1, 2\} = 2$ elements	$(1, 1, 2)$ has 3 entries
Answers the question	which things?	which things, in which slots?

Watch out: two tuples of different lengths are never equal

$(1, 2)$ and $(1, 2, 0)$ are not equal, even though the extra entry is 0: one has two slots and the other has three, so there is no way to match them entry by entry. In data terms, a record with a missing column is not the same as a record with a 0 in that column.

8.3 Why Tuples Matter Here

Tuples are the bridge to the rest of the course.

- **Coordinates:** every point on a graph is a tuple.
- **Relations and functions (Unit 1-3):** a relation is literally a *set of ordered pairs*, each pair written as (input, output).
- **Vectors (Unit 2):** a vector in $\mathbb{R}^n$ is an n -tuple of numbers. Here $\mathbb{R}^n$ means the set of all n -tuples of real numbers; for instance, $\mathbb{R}^2$ is all pairs (x, y) , the plane. In Unit 2 we will write a vector's entries stacked in a column rather than across a line, but it carries the same information: n numbers in a fixed order.
- **Data:** one row of a data table, one patient's measurements, is a tuple, with the table's columns fixing which slot holds which measurement. A whole dataset is a set of tuples.

Each bullet in one concrete example:

- **Coordinates:** the point $(3, 4)$ in the plane, three right and four up.
- **A relation:** “patient and age” for the first three study patients is the set of ordered pairs $\{(P1, 58), (P2, 71), (P3, 34)\}$. It is a *set* (the order in which the pairs are listed does not matter) of *ordered pairs* (inside each pair, the patient comes first and the age second). Unit 1-3 builds functions on exactly this idea.
- **A vector:** $(3, 4)$ again, now read as an arrow from the origin to that point, an element of $\mathbb{R}^2$. Unit 2 will write it as a column, but the two entries and their order are the same.

- **A dataset:** the whole study table is the set of its twelve row-tuples, running from

(P1, 58, F, yes, yes, no, no) to (P12, 31, F, yes, yes, no, yes).

Watch out: a set of rows needs a way to tell rows apart

Because a set ignores repeats, two identical rows would collapse into one if the dataset were literally a set. Our study table avoids the problem because the first column, the patient ID, is different in every row, so no two row-tuples are equal. A table without an ID column (say, a list of lab readings where two readings happened to have identical values) should be thought of as a *list* of tuples, in which repeats count. Removing duplicate rows is then a deliberate choice, not something that happens by itself.

The takeaway

A **set** answers “which things?” A **tuple** answers “which things, in which slots?” Nearly all of data science is built out of the two.

8.4 Extra Practice

Exercise 8.1

Equal or not? Give a reason for each.

1. $(2, 3)$ and $(3, 2)$
2. $\{2, 3\}$ and $\{3, 2\}$
3. $(1, 2, 2)$ and $(1, 2)$
4. $\{1, 2, 2\}$ and $\{1, 2\}$
5. $(4, 0)$ and $(4, 0, 0)$

Solution to Exercise 8.1

1. **Not equal.** The first entries differ (2 versus 3), and for tuples that alone settles it.
2. **Equal.** Both sets contain exactly 2 and 3; order does not matter in a set.
3. **Not equal.** The first has three entries, the second two; tuples of different lengths are never equal.
4. **Equal.** The repeated 2 does not count in a set, so $\{1, 2, 2\} = \{1, 2\}$.
5. **Not equal.** Different lengths again, two entries versus three, even though the extra entry is 0.

Items 1 and 2, and items 3 and 4, show the two ways sets and tuples differ: order, and repeats.

Exercise 8.2

Solve $(2x, y - 1) = (6, x)$ for x and y .

Solution to Exercise 8.2

Match entry by entry: $2x = 6$ and $y - 1 = x$. The first gives $x = 3$. Substituting into the second, $y - 1 = 3$, so $y = 4$. Check: $(2 \times 3, 4 - 1) = (6, 3)$, and the right side is $(6, x) = (6, 3)$. They agree.

Exercise 8.3

Using the study table in Section 1.3:

1. Write patient 10's record as a 7-tuple, in the table's column order.
2. Write the relation "patient and sex" for patients 1 to 4 as a set of ordered pairs.
3. Would the relation in part 2 change if you listed its pairs in a different order? Would it change if you wrote each pair as (sex, patient)?

Solution to Exercise 8.3

1. (P10, 67, F, yes, no, yes, no): aged 67, female, diabetic, not hypertensive, obese, not pregnant.
2. $\{(P1, F), (P2, M), (P3, F), (P4, M)\}$.
3. Listing the pairs in a different order changes nothing: the relation is a *set* of pairs, and sets ignore order. Writing each pair as (sex, patient) *does* change it: (F, P1) is a different ordered pair from (P1, F), so the result is a different set of pairs (it describes the same facts, read in the other direction). Order matters inside each pair, but not among the pairs.

9 Review

What we covered

1. A **set** is a collection of elements; $\{x \mid \dots\}$ describes one by a rule.
2. $A \subseteq B$ means every element of A is in B ; $\emptyset$ is a subset of everything.
3. $A \cup B$, $A \cap B$, $A \setminus B$, $\bar{A}$ are the set versions of OR, AND, AND NOT, NOT, and **De Morgan's laws** carry over.
4. **Venn diagrams** organize the regions; $|A \cup B| = |A| + |B| - |A \cap B|$ counts them.
5. $\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$; rational decimals terminate or repeat, irrational ones do not.
6. $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ are all **countable** ($\aleph_0$); Cantor's diagonal shows $\mathbb{R}$ is strictly bigger.
7. **Intervals** are stretches of the real line; brackets record the endpoints.
8. **Tuples** are collections where order matters: points, data rows, vectors.

The mistakes to watch for

- Confusing $\in$ (an element in a set) with $\subseteq$ (a set inside a set), and $\emptyset$ with $\{\emptyset\}$.
- Counting repeats, or counting rows instead of distinct values, when finding a cardinality.
- Reading an English “and” as AND when the speaker is joining two lists (OR).
- Adding $|A| + |B|$ for “ A or B ” without subtracting the overlap; using a pairwise total as a “two-way only” region in a three-set diagram.
- Pushing a complement bar inward without flipping $\cup$ and $\cap$ (De Morgan).
- Taking a complement without saying what the universe is.
- Classifying a number by its appearance ($\sqrt{9}$, $\frac{8}{2}$) instead of its value.
- Getting an interval’s endpoints wrong: a union includes a shared endpoint if either interval does, an intersection only if both do, and a difference flips the removed interval’s brackets.
- Treating ordered data as a set, which loses both the order and the repeats.

Notebooks:

- U1-2_Sets-1_Operations.ipynb
- U1-2_Sets-2_VennDiagrams.ipynb
- U1-2_Sets-3_NumberSystems.ipynb
- U1-2_Sets-4_Intervals.ipynb

Previous: Unit 1-1, **Logic**.

Next: Unit 1-3, **Functions**.